TCS NQT · ADVANCED QUANT

Advanced Quant Zero to Hero

Every topic from basics to the hardest question: concepts, formulas, shortcuts, how questions are asked, solved examples and graded practice.

15	114	149	23
chapters, zero to hero	solved examples	practice questions	diagrams and charts

ONLINE LEARNING MOCK TEST SERIES · ATC DEVELOPERS

About this guide

This guide covers every Advanced Quantitative topic that can appear in the TCS NQT Advanced section, starting from the basic idea and ending at exam-level difficulty. Each chapter shows how the questions are framed, the formulas to remember, the fastest method and the mistakes that cost marks.

How each chapter is built

- **Idea** in plain words, then a diagram.
- **Formulas and rules** in the yellow box.
- **How the questions appear** in the exam.
- **Solved examples** with full steps.
- **Shortcuts and mistakes** to avoid.
- **Practice set** (Basic, Medium, Advanced) with answers at the end.

How to study

- Read the chapters in order the first time; the Must-do labels give the biggest return.
- Do the Basic questions after each chapter, the Medium ones the next day.
- Time the Advanced ones: about 2 minutes each.
- Finish with the mini test and keep the formula sheet for the last day.

Contents

1. How the Advanced Quant Section Works — Foundation — 3
2. Number Theory: Factors, HCF, LCM, Divisibility — Must-do — 6
3. Remainders, Cyclicity, Number Systems — Must-do — 9
4. Algebra and Equations — Important — 13
5. Sequences, Series, Logs, Indices, Surds — Important — 16
6. Permutation and Combination — Must-do — 20
7. Probability — Must-do — 24
8. Set Theory and Venn Diagrams — Important — 27
9. Data Interpretation — Must-do — 30
10. Time and Work, Pipes and Cisterns — Must-do — 33
11. Time, Speed and Distance (Advanced) — Must-do — 37
12. Percentages, P&L, Mixtures, Partnership — Must-do — 40
13. Simple and Compound Interest — Important — 44
14. Mensuration and Geometry — Important — 47
15. Calculation Tricks, Mini Test, Formula Sheet — Final — 51

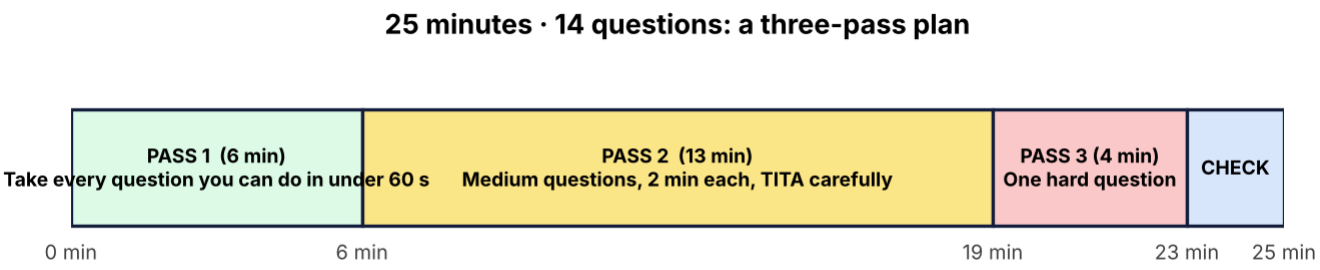
Honest note

- Every number in the solved examples and in the practice answers was calculated by a program while this guide was made, and puzzle answers were found by exhaustive search where possible.
- TCS does not publish its question weightage. Priority labels are our preparation guidance.

ADVANCED QUANT · START HERE

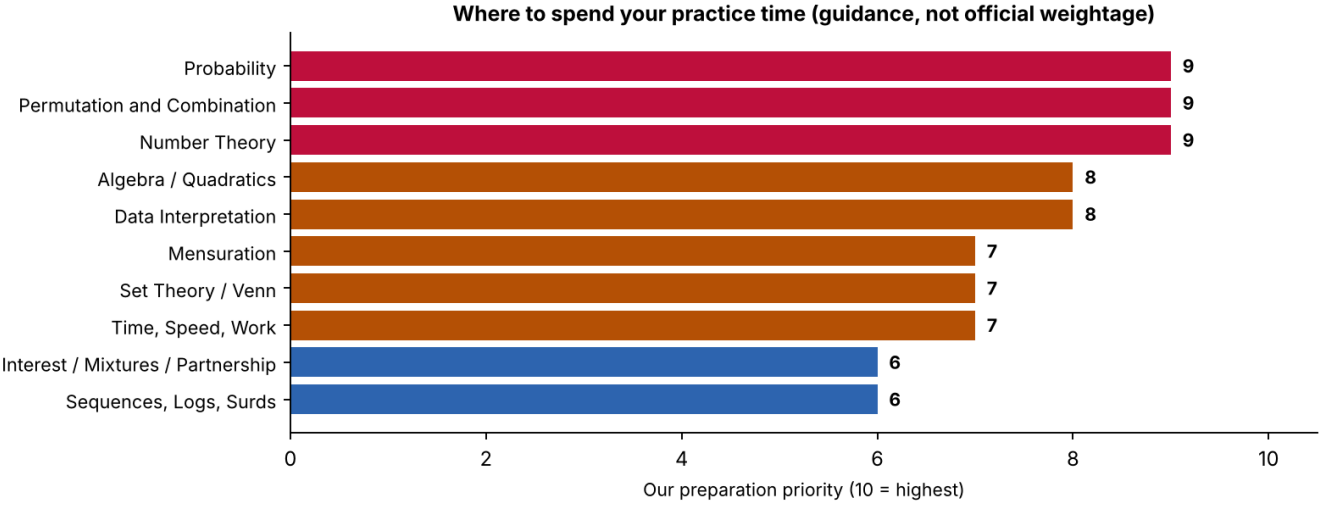
01 How the Advanced Quant Section Works

The Advanced section of the NQT is for candidates aiming at the higher TCS profiles (Digital and Prime). In the Advanced Quantitative and Reasoning block you get **14 questions in 25 minutes**. The quantitative questions are harder than the Foundation Numerical section: longer wording, two or three steps, and many **typed-answer (TITA)** questions where there are no options to fall back on. This guide covers the quantitative side from zero to a high level; the companion Advanced Reasoning guide covers the logic side.



A three-pass plan for 14 questions in 25 minutes.

Feature	What to expect
Questions and time	14 questions, 25 minutes, so about 107 seconds each.
Question types	MCQ with 4 options, and typed numerical answers (TITA).
Marking	1 mark each. No negative marking has been reported for this pattern, so attempt everything.
Style	Word problems with two or three layers, data tables, probability and counting, number theory with a twist, equations and mixed reasoning puzzles.
What is rewarded	Choosing the right method in the first 15 seconds, then clean and accurate arithmetic.

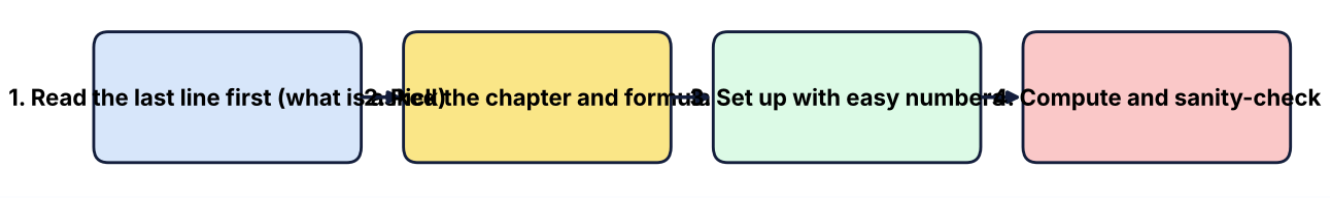


Priority order for preparation. Probability, counting and number theory give the most reliable marks.

The five habits of high scorers

- **Classify first.** Spend 10 seconds deciding: which chapter, which formula, which shortcut.
- **Convert to easy numbers.** Replace percentages by fractions (12.5% = 1/8), use LCM for work problems, use ratios instead of actual values.
- **Estimate before you calculate.** If options differ by a lot, rough arithmetic is enough.
- **For TITA, double-check the unit and the form** (integer, decimal, fraction) because there is no option to confirm against.
- **Skip without guilt.** A question that is not moving in 2 minutes should be marked and left.

How a question is solved: the 4 steps



Read the question twice only if it is a word problem with many conditions.

How many positive factors does 210 have?

16

210 = 2 × 3 × 5 × 7, so the number of factors = (1+1)(1+1)(1+1)(1+1) = 16. This is a Number Theory 15-second question: classify, apply the formula, done.

A shopkeeper marks an article 25% above cost and gives a 20% discount. What is his profit percent?

0%

Cost = 100, marked = 125, selling = 125 × 0.8 = 100. Profit = 0%. Using multipliers: 1.25 × 0.8 = 1.

Shortcuts and tricks

- Do the chapter you love first in Pass 1; confidence saves time later.
- Keep the last 2 minutes only for re-checking TITA answers.
- Write the key number on the sheet; do not hold it in your head.
- Learn the tables in the last chapter (squares to 30, cubes to 15, fractions to percent) before anything else.

Common mistakes

- Starting the very first question with a long calculation without checking its type.
- Leaving a TITA blank: there is no negative marking, so write your best answer.
- Spending 4 minutes on a 3-mark-worth puzzle while easy questions stay untouched.

Practice set: Warm-up (time yourself: 90 seconds each)

1. How many positive factors does 360 have? Basic

2. A number is increased by 20% and then decreased by 20%. Net change in percent (negative = decrease)? Basic

3.	What is 12.5% of 640?	Basic
4.	If the ratio of A to B is 3:5 and A = 27, what is B?	Basic
5.	The average of 5 numbers is 24. If one number 30 is removed, the average of the remaining numbers is?	Medium
6.	How many numbers from 1 to 100 are divisible by 3 or 5?	Medium

ADVANCED QUANT · NUMBER THEORY · MUST-DO

02 Number Theory: Factors, HCF, LCM and Divisibility

Number theory is the most dependable scoring area of the advanced quant section. Questions look scary (huge powers, many conditions) but each is solved by a **short formula**: count factors from the prime factorisation, use HCF and LCM for "greatest" and "least" numbers, use the unit-digit cycle for big powers. Learn the formulas and the 20-second techniques below.

Divisibility tests you must know

Divisor	Test	Example
2, 4, 8	last 1, 2, 3 digits divisible by 2, 4, 8	5 7 6 → 576/8 = 72 ✓
3, 9	digit sum divisible by 3, 9	4 8 6 → 18 ✓ for 3 and 9
5, 25	last digit 0/5; last two digits 00/25/50/75	1 2 7 5 ✓ for 25
6	divisible by 2 and 3	2 1 6 ✓
7	drop last digit, subtract twice it	343: 34 − 6 = 28 ✓
11	(sum odd places) − (sum even places) = 0 or 11k 9 1 8	→ (9+8)−1 = 16 × ; 2 7 5 6 3: (2+5+3)−(7+6)=−3 ×
13	add 4 × last digit to the rest	91: 9 + 4 = 13 ✓

Divisibility tests: the 7, 11 and 13 tests save the most time.

Factors ($N = p^a q^b r^c \ldots$)

- Number of factors = **(a+1)(b+1)(c+1)….**
- Sum of factors = $(p^{a+1}-1)/(p-1) \times (q^{b+1}-1)/(q-1) \times \ldots$
- Product of all factors = N^(d/2), where d is the number of factors.
- Number of **odd** factors: ignore the power of 2. Number of **even** factors = total − odd.
- A number is a perfect square exactly when every power is even, i.e. it has an **odd** number of factors.
- Ways to write N as a product of two factors = d/2 (or (d+1)/2 for a perfect square when order does not matter).
- Numbers below N that are coprime to N (Euler): $\varphi(N) = N \times (1-1/p)(1-1/q)\ldots$

How many factors does 7200 have? How many of them are odd? How many are perfect squares?

54; 9; 12

$7200 = 2^5 \times 3^2 \times 5^2$. Factors = 6 × 3 × 3 = 54. Odd factors ignore the 2s: 3 × 3 = 9. Square factors take even powers: (0,2,4) × (0,2) × (0,2) = 3 × 2 × 2 = 12.

Find the sum of all positive factors of 360 and the count of numbers up to 360 that are coprime to 360.

1170; 96

360 = $2^3 \times 3^2 \times 5$. Sum = $(2^4-1)/1 \times (3^3-1)/2 \times (5^2-1)/4 = 15 \times 13 \times 6 = 1170$. $\varphi(360) = 360 \times ½ \times ⅔ \times ⅘ = 96$.

What is the smallest positive integer that has exactly 15 factors?

144

15 = 15 × 1 or 5 × 3. The smallest is with 5 × 3: $2^4 \times 3^2 = 144$. (Always put the largest power on the smallest prime.)

HCF and LCM: the four standard question types

Rules

- **HCF × LCM = product of the two numbers** (only for two numbers).
- Greatest number that divides a, b, c leaving remainders r_1, r_2, r_3: **HCF(a − r_1, b − r_2, c − r_3)**.
- Least number that leaves the **same remainder r** when divided by x, y, z: **LCM(x, y, z) + r**.
- Least number that is **r short** of a multiple (x − a = y − b = k): **LCM − k**.
- Greatest number that divides a, b, c leaving the **same remainder**: HCF of the differences (a−b, b−c).
- LCM of fractions = LCM(numerators)/HCF(denominators); HCF of fractions = HCF(numerators)/LCM(denominators).
- Bells, traffic lights and runners meeting again: answer is the **LCM of their time periods**.

Find the greatest number that divides 1657 and 2037 leaving remainders 6 and 5 respectively.

127

Required number = HCF(1657−6, 2037−5) = HCF(1651, 2032) = 127.

Find the least number which leaves remainder 5 when divided by 12, 15 and 20.

65

LCM(12, 15, 20) = 60, so the number is 60 + 5 = 65.

Three bells ring at intervals of 18, 24 and 30 minutes and ring together at 8:00 a.m. At what time will they next ring together?

2:00 p.m.

LCM(18, 24, 30) = 360 minutes = 6 hours. 8:00 a.m. + 6 hours = 2:00 p.m.

Trailing zeros and the power of a prime in n!

- Trailing zeros in n! = $\lfloor n/5 \rfloor + \lfloor n/25 \rfloor + \lfloor n/125 \rfloor + \ldots$
- Highest power of a prime p in n! = $\lfloor n/p \rfloor + \lfloor n/p^2 \rfloor + \lfloor n/p^3 \rfloor + \ldots$
- Trailing zeros of a product: count the pairs (2, 5) in its prime factorisation.

How many trailing zeros are in 100!? In 250!?

24; 62

100!: 20 + 4 = 24. 250!: 50 + 10 + 2 = 62.

What is the highest power of 2 that divides 100!?

97

⌊100/2⌋ + ⌊100/4⌋ + ⌊100/8⌋ + ⌊100/16⌋ + ⌊100/32⌋ + ⌊100/64⌋ = 50 + 25 + 12 + 6 + 3 + 1 = 97.

How these questions appear in the exam

Pattern	Typical wording	Method
Count factors	"How many divisors of 2400 are multiples of 10?"	Prime factorise, restrict the powers.
Smallest number with k factors	"Smallest number with 12 factors"	Largest power on the smallest prime.
HCF/LCM word problem	Bells, tiles, ropes, runners, greatest length measuring	HCF for "greatest", LCM for "least / together".
Remainders with HCF/LCM	"Divides 100 and 140 leaving remainders..."	Subtract remainders, then HCF.
Factorial zeros	"Number of zeros at the end of 200!"	Divide by 5, 25, 125.

Shortcuts and tricks

- A multiple of 10 needs at least one 2 and one 5: count factors with powers restricted to ≥1 of each.
- Two numbers with HCF h are h·a and h·b with a, b coprime. Use this to build the numbers.
- For "greatest number ... same remainder" always use the difference of the numbers.
- Test your answer: does it really leave the stated remainders?

Common mistakes

- Using HCF × LCM = product for three numbers.
- Forgetting to include 1 and N itself when counting factors.
- Subtracting the remainder in an LCM problem (it is added) or adding it in an HCF problem (it is subtracted).

Practice set: Factors, HCF and LCM practice

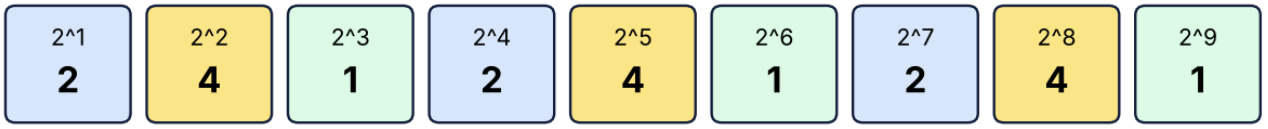
1. How many factors does 2520 have? — Basic
2. What is the sum of the factors of 100? — Basic
3. Find the HCF of 36 and 84. — Basic
4. Find the LCM of 12, 18 and 27. — Basic
5. How many trailing zeros does 150! have? — Medium
6. Find the least number which, when divided by 6, 8 and 12, leaves remainder 3 each time. — Medium
7. How many divisors of 3600 are perfect squares? — Medium
8. Find the greatest number that divides 245 and 1029 leaving remainder 5 in each case. — Advanced
9. How many integers from 1 to 500 are coprime to 500? — Advanced
10. Find the smallest number having exactly 18 factors. — Advanced

ADVANCED QUANT · NUMBER THEORY · MUST-DO

03 Remainders, Cyclicity and Number Systems

Big powers and big factorials cannot be computed, but their **remainders and last digits repeat in short cycles**. This chapter shows how to find the unit digit, last two digits and remainders of huge numbers, how to combine remainders (Chinese Remainder idea), and how number systems and digit sums work.

Remainder pattern of 2^n when divided by 7 (repeats every 3 steps)



1 → 2 → 4 → 1 → 2 → 4 ... so 2^100: 100 mod 3 = 1 → remainder 2

Find the cycle length, reduce the power modulo that length, read the remainder.

Remainders repeat. Always look for the cycle first.

Unit digit cycles of powers

Last digit	Cycle of unit digits	Length
2	2 → 4 → 8 → 6	4
3	3 → 9 → 7 → 1	4
4	4 → 6	2
7	7 → 9 → 3 → 1	4
8	8 → 4 → 2 → 6	4
9	9 → 1	2
5	5	1
6	6	1

Unit-digit cycles. For digits 0, 1, 5, 6 the unit digit never changes.

Rules

- **Unit digit of a^b:** take the last digit of a, find its cycle (length 1, 2 or 4), reduce b modulo the cycle length (use the full cycle when the remainder is 0).
- **Remainder rules:** (a×b) mod n = ((a mod n)(b mod n)) mod n, and the same for + and −.
- **Fermat:** if p is prime and p does not divide a, then a^(p−1) ≡ 1 (mod p).
- **Euler:** if gcd(a, n) = 1, then a^φ(n) ≡ 1 (mod n). Use it with φ(100) = 40 for last two digits.
- **Remainder with a negative number:** 24 mod 25 = −1. So 24^n mod 25 = $(-1)^n$.
- **Remainder theorem:** the remainder of p(x) divided by (x − a) is p(a).
- **$(a^n + b^n)$** is divisible by (a + b) when n is odd. **$(a^n - b^n)$** is divisible by (a − b) always and by (a + b) when n is even.
- **Digit sum / digital root:** a number leaves the same remainder as its digit sum when divided by 9 (and by 3).

Find the unit digit of 7^2023.

3

Cycle of 7: 7, 9, 3, 1 (length 4). 2023 mod 4 = 3, so the digit is the 3rd in the cycle: 3.

Find the last two digits of 3^200.

1

φ(100) = 40 and gcd(3, 100) = 1, so $3^{40} \equiv 1 \pmod{100}$. 200 = 5 × 40, hence $3^{200} \equiv 1$ → last two digits are 01. (Answer entered as 1.)

What is the remainder when 2^100 is divided by 7?

2

$2^3 = 8 \equiv 1 \pmod 7$. 100 = 3 × 33 + 1, so $2^{100} \equiv 2^1 = 2$.

What is the remainder when 17^2024 is divided by 9?

1

$17 \equiv 8 \equiv -1 \pmod 9$. So $17^{2024} \equiv (-1)^{2024} = 1$, remainder 1.

What is the remainder when 5^100 is divided by 13?

1

$5^4 = 625 = 13 \times 48 + 1$, so $5^4 \equiv 1 \pmod{13}$. 100 = 4 × 25 → remainder 1.

Combining remainders (Chinese Remainder idea)

Find a number that leaves different remainders with different divisors. Build it step by step: start with the largest modulus, then add its multiples until the next condition is met.

Find the smallest positive number that leaves remainder 3 when divided by 7, remainder 5 when divided by 11 and remainder 2 when divided by 13.

808

The numbers satisfying the 13-condition are 2, 15, 28, 41, 54, 67, 80, ... Among those leaving remainder 5 with 11 come 80 and then every 143. Among those with remainder 3 with 7 the smallest is 808. It repeats every 7 × 11 × 13 = 1001 numbers.

Number systems and digit tricks

Rules

- Convert from base b to decimal by the place values: $1011_2 = 8 + 0 + 2 + 1 = 11$.
- Convert from decimal to base b by repeated division, reading remainders from the bottom up.
- **Digital root** of $n = 1 + (n - 1) \bmod 9$ (for $n > 0$). The digital root of a product is the digital root of the product of the roots.
- A number and its reversal differ by a multiple of 9. For a two-digit number ab, $ab - ba = 9(a - b)$.
- The sum of the digits of numbers from 1 to 99 together is 900. The sum of all numbers from 1 to n is $n(n+1)/2$.
- Squares end only in 0, 1, 4, 5, 6, 9. Squares of numbers ending in 5: $(...5)^2 = n(n+1)$ followed by 25.

Convert 345 to binary and to octal.

101011001 ; 531

Repeated division: $345 = 101011001_2 = 531_8$. Check: int("101011001", 2) = 345.

Find the sum of the digits of all the numbers from 1 to 100.

901

Numbers 0 to 99 pair as (k, 99 − k) with digit sum 18 each; 50 pairs → 900. Add 100 → digit sum 1, giving 901.

Find the remainder when 123456789 × 987654321 is divided by 9.

0

Digit sum of 123456789 is 45 → 0 (mod 9). So the product leaves remainder 0.

How these questions appear

Pattern	Example wording	Method
Unit digit	“Unit digit of 13^77 + 27^45”	Find each cycle, add last digits.
Last two digits	“Last two digits of 7^99”	Euler with $\varphi(100) = 40$, or use $7^4 = 2401 \equiv 01$.
Remainder of a power	“Remainder when 3^101 is divided by 11”	Find the cycle of remainders or use Fermat.
Remainder with two divisors	“Number leaves 2 and 3 on dividing by 5 and 7”	CRT: build from the largest modulus.
Number system	“$(213)_4 + (132)_4$ in base 4”	Convert, add, convert back.
Digital root	“Remainder when the product is divided by 9”	Use digit sums.

Shortcuts and tricks

- Last two digits of any number ending in 01 stay 01 for all powers; numbers ending in 25 stay 25; 76 stays 76.
- $7^4 = 2401 \Rightarrow$ last two digits of 7^n repeat every 4: 07, 49, 43, 01.
- To check a remainder answer, re-compute with a small power pattern.
- For "remainder when divided by 7, 11, 13" numbers, note 1001 = 7 × 11 × 13: a six-digit number abcabc is always divisible by 1001.

Common mistakes

- Using the cycle length when the exponent remainder is 0 (you need the last number of the cycle, not the first).
- Applying Euler's theorem when gcd(a, n) ≠ 1.
- Treating remainders as negative numbers in the final answer (give the positive remainder).

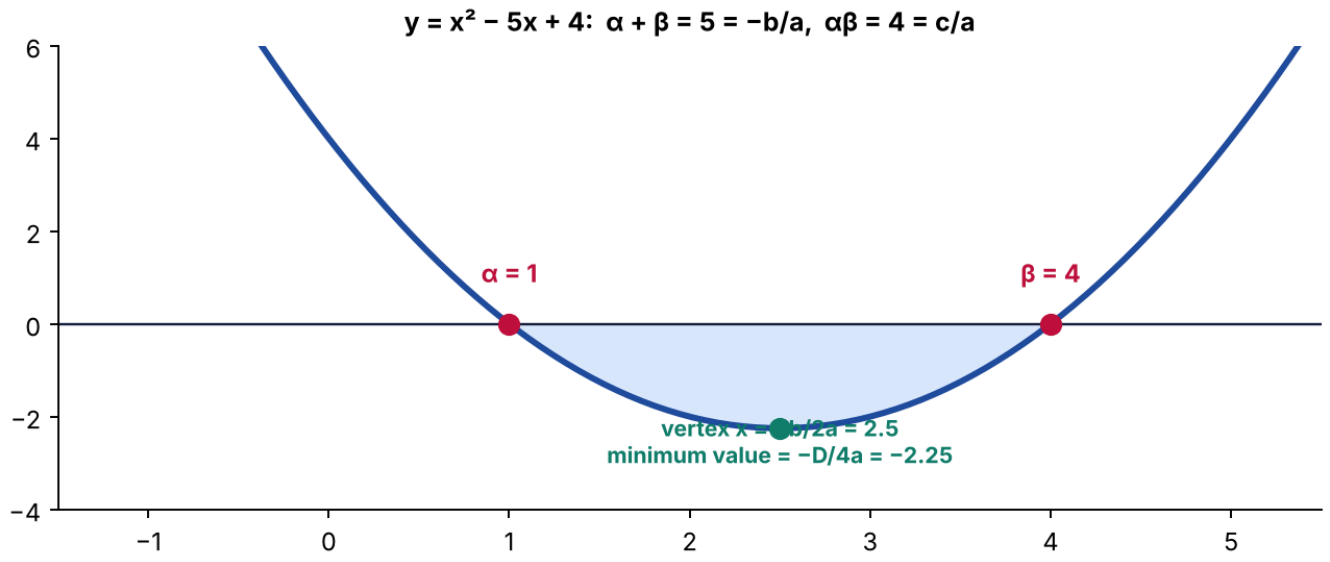
Practice set: Remainders and number systems practice

1.	Unit digit of 3^77?	Basic
2.	Unit digit of 2^99 + 3^99?	Basic
3.	Remainder when 2^50 is divided by 7?	Medium
4.	Remainder when 7^100 is divided by 5?	Medium
5.	Last two digits of 7^99?	Medium
6.	Convert 255 to binary.	Basic
7.	Remainder when 100! + 5 is divided by 7? (hint: 100! is a multiple of 7)	Medium
8.	Smallest positive number leaving remainders 1, 2, 3 when divided by 3, 4, 5 respectively.	Advanced
9.	Remainder when 3^200 is divided by 17?	Advanced
10.	Last digit of 13^13 × 17^17?	Advanced

ADVANCED QUANT · ALGEBRA · MUST-DO

04 Algebra: Equations, Quadratics and Inequalities

Algebra questions in the advanced section are rarely “solve for x”. They ask for **expressions in the roots** ($\alpha^2 + \beta^2$, $\alpha^3 + \beta^3$), for the **number of integer solutions**, for a **maximum or minimum value**, or for an unknown found from a pair of equations. The shortcut is nearly always a **symmetric identity**: express the wanted quantity through the sum and the product of the roots.



A quadratic and its key numbers: roots, sum, product and vertex.

Quadratic $ax^2 + bx + c = 0$

- Roots α, β: **$\alpha + \beta = -b/a$**, **$\alpha\beta = c/a$**. Discriminant $D = b^2 - 4ac$.
- $D > 0$: two real roots; $D = 0$: equal roots; $D < 0$: no real roots. Roots are rational only if D is a perfect square.
- **$\alpha^2 + \beta^2 = (\alpha+\beta)^2 - 2\alpha\beta$**; **$(\alpha-\beta)^2 = (\alpha+\beta)^2 - 4\alpha\beta$**.
- **$\alpha^3 + \beta^3 = (\alpha+\beta)^3 - 3\alpha\beta(\alpha+\beta)$**; $1/\alpha + 1/\beta = (\alpha+\beta)/\alpha\beta$.
- New equation with roots $\alpha+k$, $\beta+k$: replace x by $(x - k)$. With roots $1/\alpha$, $1/\beta$: reverse the coefficients.
- Minimum of $ax^2 + bx + c$ ($a > 0$) is at $x = -b/2a$ with value **$-D/4a$**. Maximum for $a < 0$.

Identities and inequalities

- $(a+b)^2 = a^2+2ab+b^2$; $a^3+b^3 = (a+b)(a^2-ab+b^2)$; $a^3+b^3+c^3-3abc = (a+b+c)(a^2+b^2+c^2-ab-bc-ca)$.
- If $a + b + c = 0$ then $a^3 + b^3 + c^3 = 3abc$.
- **If $x + 1/x = k$:** $x^2 + 1/x^2 = k^2 - 2$ and $x^3 + 1/x^3 = k^3 - 3k$.
- **AM ≥ GM:** for positive numbers, $a + b \geq 2\sqrt{(ab)}$. If the sum is fixed the product is maximum when the numbers are equal; if the product is fixed the sum is minimum when equal.
- **Modulus:** $|x - a| < b$ means $a - b < x < a + b$. $|x - a| > b$ means $x < a - b$ or $x > a + b$.
- Floor function: $\lfloor x \rfloor \leq x < \lfloor x \rfloor + 1$.
- **Remainder theorem:** $p(x) \div (x - a)$ leaves $p(a)$. **Factor theorem:** $(x - a)$ is a factor if $p(a) = 0$.

If α and β are the roots of $x^2 - 7x + 10 = 0$, find $\alpha^2 + \beta^2$, $\alpha^3 + \beta^3$ and $(\alpha - \beta)^2$.

29; 133; 9

$\alpha + \beta = 7$, $\alpha\beta = 10$. $\alpha^2 + \beta^2 = 7^2 - 2\times10= 29$. $\alpha^3 + \beta^3 = 7^3 - 3\times10\times7 = 133$. $(\alpha - \beta)^2 = 7^2 - 4\times10 = 9$.

If $x + 1/x = 5$, find $x^3 + 1/x^3$ and $x^4 + 1/x^4$.

110; 527

$x^2 + 1/x^2 = 23$. $x^3 + 1/x^3 = 5^3 - 3\times5 = 110$. $x^4 + 1/x^4 = (23)^2 - 2 = 527$.

How many integers x satisfy $x^2 - 5x - 14 < 0$?

8

Factorise: $(x - 7)(x + 2) < 0$, so $-2 < x < 7$. Integers: $-1, 0, 1, \ldots, 6 \rightarrow 8$.

If x and y are positive integers with $x + y = 31$, find the maximum value of xy.

240

For a fixed sum the product is maximum when the numbers are as close as possible: $15 \times 16 = 240$.

Find the remainder when $x^3 - 6x^2 + 11x + 4$ is divided by $(x - 2)$.

10

By the remainder theorem put $x = 2$: $8 - 24 + 22 + 4 = 10$.

Word problems that become equations

The sum of a number and its reciprocal is 10/3. Find the number(s).

3 or 1/3

$x + 1/x = 10/3 \Rightarrow 3x^2 - 10x + 3 = 0 \Rightarrow (3x - 1)(x - 3) = 0$.

Pens cost ₹3 and notebooks cost ₹2 each. Ravi bought some of each for exactly ₹60 and bought more pens than notebooks, at least one of each. In how many ways is this possible?

3

$3x + 2y = 60$ with $x > y \geq 1$. Since 60 and $2y$ are even, $3x$ is even, so x is even: $x = 2, 4, 6, \ldots$ and $y = (60 - 3x)/2$ must be a positive integer smaller than x. Listing the pairs gives 3 solutions.

How these questions appear

Pattern	Wording	Method
Symmetric in the roots	"Find $\alpha^3 + \beta^3$"	Use sum and product of roots.
Equation from conditions	"Roots are consecutive integers with product 56"	Set up S and P.
Integer solutions	"Number of integers with $\lvert x - 2\rvert \leq 7$"	Write the interval and count.
Max/min of an expression	"Maximum of xy with $x + y = 40$"	AM-GM or vertex of the parabola.
Linear Diophantine	"Ways to pay ₹60 with 3- and 2-rupee coins"	Parity and list the pairs.
Polynomial remainder	"Remainder when $p(x) \div (x - 2)$"	Substitute $x = 2$.

Shortcuts and tricks

- If the question gives S and P, never solve for the roots.
- To count integer solutions, convert the inequality to an interval first; the endpoints matter for ≤ and <.
- For "maximum value of a product with a fixed sum", make the numbers equal; for integers make them nearly equal.
- For equations with unknown integers, use parity or divisibility to cut the cases.

Common mistakes

- Forgetting that D must be a perfect square for rational roots.
- Including the endpoints for strict inequalities.
- Using AM-GM when the numbers can be negative.

Practice set: Algebra practice

1.	If $\alpha + \beta = 6$ and $\alpha\beta = 5$, find $\alpha^2 + \beta^2$.	Basic
2.	If $x + 1/x = 3$, find $x^2 + 1/x^2$.	Basic
3.	Find the sum of the roots of $2x^2 - 9x + 4 = 0$.	Basic
4.	How many integers satisfy $\lvert x - 4 \rvert < 6$?	Medium
5.	Minimum value of $x^2 - 8x + 20$?	Medium
6.	Find the remainder when $x^4 - 3x^2 + 5$ is divided by $x + 1$.	Medium
7.	If $a + b + c = 0$, find $(a^3 + b^3 + c^3)/abc$.	Medium
8.	For positive x, the minimum value of $x + 16/x$?	Advanced
9.	How many positive integer pairs (x, y) satisfy $4x + 5y = 100$?	Advanced
10.	If $x^2 + 1/x^2 = 14$ and $x > 0$, find $x + 1/x$.	Advanced

ADVANCED QUANT · ALGEBRA · IMPORTANT

05 Sequences, Series, Logarithms, Indices and Surds

This chapter collects the formula-driven topics: arithmetic and geometric series, special sums, logarithms, indices and surds. Questions are short but need exact formulas. Many are solved in one line once you recognise the pattern, for example a **telescoping** product of logarithms or a series whose sum is a perfect square.

Series and logarithm formulas

AP: $a_n = a + (n-1)d,\ S_n = \frac{n}{2}[2a + (n-1)d]$

GP: $a_n = ar^{n-1},\ S_n = \frac{a(r^n - 1)}{r - 1}$

Infinite GP ($|r| < 1$): $S_\infty = \frac{a}{1 - r}$

HP: reciprocals form an AP

$1 + 2 + \cdots + n = \frac{n(n+1)}{2}$

$1^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}$

$1^3 + \cdots + n^3 = \left[\frac{n(n+1)}{2}\right]^2$

Sum of first n odd numbers $= n^2$

$\log_a(xy) = \log_a x + \log_a y$

$\log_a(x/y) = \log_a x - \log_a y$

$\log_a x^n = n \log_a x,\ \log_a b = \frac{\log b}{\log a}$

$a^{\log_a x} = x,\ \log_a b \times \log_b a = 1$

The formulas to memorise for series and logarithms.

Indices and surds

$a^m \times a^n = a^{m+n}$

$(a^m)^n = a^{mn}$

$a^0 = 1,\ a^{-n} = \frac{1}{a^n}$

$(\sqrt{a} + \sqrt{b})(\sqrt{a} - \sqrt{b}) = a - b$

$\frac{1}{\sqrt{a} + \sqrt{b}} = \frac{\sqrt{a} - \sqrt{b}}{a - b}$

$\sqrt{a + b + 2\sqrt{ab}} = \sqrt{a} + \sqrt{b}$

Indices and surds.

The first term of an AP is 7 and the common difference is 4. Find the sum of the first 40 terms and the 40th term.

3400; 163

a_n = 7 + 39×4 = 163. S = 40/2 × (2×7 + 39×4) = 20 × 170 = 3400.

Find the sum of all multiples of 7 between 1 and 200.

2842

7, 14, ..., 196 form an AP with 28 terms. Sum = 28/2 × (7 + 196) = 2842.

Find the sum of the series 3 + 6 + 12 + ... up to 8 terms.

765

GP with a = 3, r = 2: $3(2^8 - 1)/(2 - 1) = 765$.

Find the sum of the infinite series 6 + 2 + 2/3 + 2/9 + ...

9

a = 6, r = 1/3: 6/(1 − 1/3) = 9.

Find the value of $1^2 + 2^2 + 3^2 + ... + 20^2$.

2870

$n(n+1)(2n+1)/6 = 20 \times 21 \times 41/6 = 2870$.

Find the value of 1/(1×2) + 1/(2×3) + 1/(3×4) + ... + 1/(19×20).

19/20

Each term = 1/i − 1/(i+1), so the sum telescopes to 1 − 1/20 = 19/20.

Telescoping tricks

- **1/(n(n+1)) = 1/n − 1/(n+1).** Also 1/(n(n+k)) = (1/k)(1/n − 1/(n+k)).
- **Product of logs:** $\log_2 3 \times \log_3 4 \times ... \times \log_{(n-1)} n = \log_2 n$.
- 1/(√n + √(n+1)) = √(n+1) − √n. The sum of such terms collapses to the last minus the first.
- Series of the type 1·2 + 2·3 + ... + n(n+1) = n(n+1)(n+2)/3.

Find the value of $\log_2 3 \times \log_3 4 \times \log_4 5 \times ... \times \log_{15} 16$.

4

The product telescopes to $\log_2 16 = 4$.

If $\log_{10} 2 = 0.3010$, find the number of digits in 2^64.

20

Digits = ⌊64 × 0.3010⌋ + 1 = ⌊19.26⌋ + 1 = 20.

If $\log_2(x) + \log_2(x - 2) = 3$, find x.

4

$\log_2[x(x - 2)] = 3 \rightarrow x(x - 2) = 8 \rightarrow x^2 - 2x - 8 = 0 \rightarrow x = 4$ (reject −2 because logs need positive numbers).

Simplify $1/(\sqrt{3}+\sqrt{2}) + 1/(\sqrt{3}-\sqrt{2})$.

$2\sqrt{3} \approx 3.464$

Rationalise: $(\sqrt{3}-\sqrt{2}) + (\sqrt{3}+\sqrt{2}) = 2\sqrt{3}$.

Find the square root of $7 + 4\sqrt{3}$.

$2 + \sqrt{3}$

Write $7 + 4\sqrt{3} = 7 + 2\sqrt{12} = 4 + 3 + 2\sqrt{(4\times3)} = (\sqrt{4}+\sqrt{3})^2 = (2+\sqrt{3})^2$.

How these questions appear

Pattern	Wording	Method
AP/GP word problem	Instalments rising by a fixed amount, bouncing ball, population	Identify a, d (or r), n; use the sum.
Special sums	$1^2 + 2^2 + \ldots$, $1\cdot2 + 2\cdot3 + \ldots$	Formulas above.
Telescoping	$1/(n(n+1))$, logs product, $\sqrt{}$ terms	Rewrite the term as a difference.
Log equations	$\log(x) + \log(x-3) = 1$	Combine, solve, reject negative arguments.
Digits in a power	"Digits in 2^100"	Use log values given.
Surd simplification	Rationalise, nested root	Conjugate or perfect square under the root.

Shortcuts and tricks

- Number of terms in an AP = (last − first)/d + 1. Count carefully: this is where most marks are lost.
- In an AP with an odd number of terms the middle term is the average, so S = n × middle term.
- For three terms in AP take (a − d, a, a + d); for GP take (a/r, a, ar).
- A log equation must be checked: every argument must stay positive.

Common mistakes

- Using n instead of n − 1 in the nth-term formula.
- Applying the infinite-GP sum when $|r| \geq 1$.
- Forgetting to reject extraneous roots of log equations.

Practice set: Sequences, logs and surds practice

1. Sum of the first 50 natural numbers? **Basic**

2. Sum of 5 + 8 + 11 + ... up to 20 terms? **Basic**

3. Find the value of $1^3 + 2^3 + \ldots + 10^3$. **Basic**

4. The 10th term of the GP 2, 6, 18, ...? **Medium**

5. Sum of $1/(1\times2) + \ldots + 1/(49\times50)$ as a fraction? **Medium**

6. If $\log_{10} 5 = 0.699$, how many digits are in 5^50? **Medium**

7. How many terms of 20 + 18 + 16 + ... add up to 110? (give the larger answer) **Advanced**

8. If x, x + 2, x + 6 are in GP, find x. **Advanced**

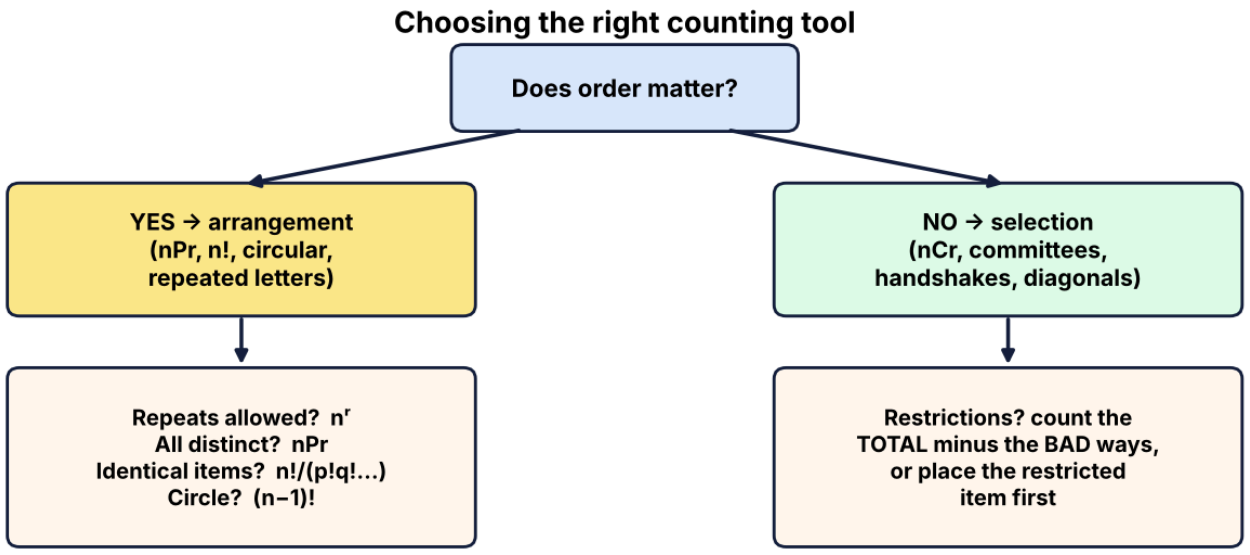
9. Value of $\log_3 81 - \log_2 8 + \log_5 125$? **Advanced**

10. Rationalise: the value of 3/(√5 − √2)? Advanced

06 Permutation and Combination

ADVANCED QUANT · COUNTING · MUST-DO

Counting questions are a favourite of the advanced section because the wording is short and the trap is the method. Decide first: **does order matter?** Then check for repeated items, restrictions and circular arrangements. A clean setup gives the answer in two lines; a wrong setup cannot be repaired by arithmetic.



Pick the tool before you pick the numbers.

Core formulas

- **nPr = n!/(n−r)!**, **nCr = n!/(r!(n−r)!)**, nCr = nC(n−r), nCr + nC(r−1) = (n+1)Cr.
- **Repeated letters:** n!/(p! q! r!...).
- **Circular arrangement** of n distinct objects: (n − 1)!. If clockwise and anticlockwise are the same (necklace): (n − 1)!/2.
- **With repetition allowed:** r positions, n choices each = n^r.
- **Distribution of identical objects** among r people: C(n + r − 1, r − 1) (anyone may get zero); each at least one: C(n − 1, r − 1).
- **Number of diagonals** of an n-gon = nC2 − n. Handshakes among n people = nC2. Triangles from n points (none three collinear) = nC3.
- **Derangements** (no item in its own place): $D_2 = 1$, $D_3 = 2$, $D_4 = 9$, $D_5 = 44$, $D_6 = 265$.
- **Paths in a grid** going only right and up across an m × n grid: C(m + n, m).

Standard question types with worked solutions

In how many ways can the letters of the word EQUATION be arranged so that all the vowels come together?

2880

Vowels E, U, A, I, O (5) and consonants Q, T, N (3). Treat the vowels as one block: 4! × 5! = 24 × 120 = 2880.

In how many ways can the letters of the word MISSISSIPPI be arranged?

34650

11!/(4! 4! 2!) because I appears 4 times, S 4 times, P 2 times.

In how many ways can 8 people be seated around a round table if two particular people must not sit together?

3600

Total = 7! = 5040. Together = 2 × 6! = 1440. Not together = 5040 − 1440 = 3600.

A committee of 5 is to be chosen from 6 men and 4 women with at least 2 women. In how many ways?

186

Cases: 2 women: 4C2 × 6C3 = 120; 3 women: 4C3 × 6C2 = 60; 4 women: 4C4 × 6C1 = 6. Total 186.

In how many ways can 10 identical chocolates be distributed among 4 children, each getting at least one?

84

Give one to each first, or use C(10 − 1, 4 − 1) = C(9, 3) = 84.

How many 4-digit numbers can be formed from the digits 1, 2, 3, 4, 5, 6 without repetition that are divisible by 5?

60

The last digit must be 5. The first three places are filled from the remaining 5 digits: 5 × 4 × 3 = 60.

How many triangles can be formed from 10 points of which 4 are collinear?

116

Choose any 3 points: 10C3 = 120, but 3 collinear points make no triangle: subtract 4C3 = 4. Answer 116.

A grid has 5 blocks east and 4 blocks north. In how many shortest ways can you walk from the bottom-left corner to the top-right corner?

126

Every path has 5 E moves and 4 N moves in some order: C(9, 4) = 126.

Five letters are placed in five addressed envelopes. In how many ways can all be placed wrongly?

44

Derangement $D_5 = 5!\,(1 - 1/1! + 1/2! - 1/3! + 1/4! - 1/5!) = 44$.

If all arrangements of the letters of the word STONE are written in dictionary order, what is the rank of STONE?

96

Alphabetical order of letters: E, N, O, S, T. Count the words before STONE position by position: first letter S: 3 smaller letters (E, N, O) × 4! = 72. Second letter T (from E, N, O, T): 3 smaller × 3! = 18. Third letter O (from E, N, O): 2 smaller × 2! = 4. Fourth letter N (from E, N): 1 smaller × 1! = 1. Fifth letter E: 0. Words before = 72 + 18 + 4 + 1 = 95, so the rank is 96. (Checked by listing all 120 arrangements.)

How these questions appear

Pattern	Wording	Method
Letters of a word	"Vowels together", "start and end with a consonant", "no two vowels together"	Block method; gap method for "no two together".
Committee selection	"At least 2 women", "a particular person included/excluded"	Cases; fix the special member; or total − bad.
Digits from a set	"Even/odd numbers, divisible by 5, greater than 5000"	Fill the restricted place first.
Circular seating	"Two persons together/apart"	(n − 1)!, treat pairs as a block.
Distribution	"Identical sweets among children"	Stars and bars.
Geometry counting	"Triangles, diagonals, rectangles in a figure"	Choose points/lines; subtract the impossible.
Grid paths	"Shortest routes"	C(m + n, m) or add paths at the corners.

Shortcuts and tricks

- **No two together:** arrange the others first, then choose gaps: for n others there are n + 1 gaps.
- "At least one" = total − none.
- If a question says "distinct" items always check for hidden repeated letters.
- Test your formula on a tiny case (3 objects) before trusting it.

Common mistakes

- Using nPr when order does not matter.
- Forgetting to multiply by the inside arrangement of a block.
- Treating circular arrangements as linear (n! instead of (n − 1)!).
- Double counting when cases overlap.

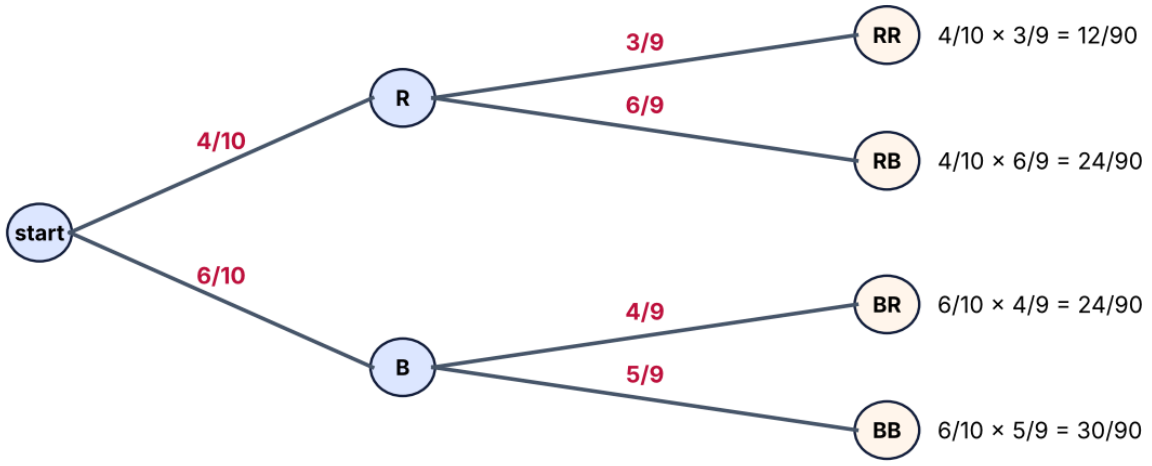
Practice set: Counting practice

1.	Number of ways to arrange the letters of BANANA?	Basic
2.	Value of 10C3?	Basic
3.	Number of handshakes among 12 people?	Basic
4.	Diagonals of a hexagon?	Basic
5.	In how many ways can 6 people sit around a round table?	Medium
6.	Number of 3-digit numbers with distinct digits formed from 0–9?	Medium
7.	A committee of 4 from 7 men and 3 women with exactly 2 women?	Medium
8.	Number of ways to distribute 12 identical balls among 3 boxes (zero allowed)?	Medium
9.	In how many ways can the letters of LEADER be arranged so that the two E's are together?	Advanced
10.	Number of derangements of 4 objects?	Advanced
11.	Shortest paths on a 6 × 3 grid (right and up)?	Advanced

07 ADVANCED QUANT · COUNTING · MUST-DO
Probability

Probability questions ask for a fraction. The safest route is **favourable outcomes ÷ total outcomes** using counting (combinations) for cards, balls and committees, and the **multiplication rule** for sequences. Advanced questions add "given that" (conditional probability), repeated trials (binomial) and expectation.

Tree for two draws without replacement from 4 red + 6 blue balls



A tree keeps without-replacement sequences organised: multiply along a branch, add across branches.

Formulas

- **P(E) = favourable / total.** $0 \le P \le 1$. P(not E) = 1 − P(E).
- **P(A or B) = P(A) + P(B) − P(A and B).** For mutually exclusive events the last term is 0.
- **Independent events:** P(A and B) = P(A) × P(B).
- **Conditional:** $P(A \mid B) = P(A \text{ and } B)/P(B)$.
- **Total probability and Bayes:** $P(A) = \Sigma P(B_i)P(A|B_i)$; $P(B_i \mid A) = P(B_i)P(A|B_i)/P(A)$.
- **Binomial:** exactly k successes in n independent trials = $nCk\ p^k\ (1-p)^{n-k}$.
- **Expectation:** E = Σ (value × probability).
- **"At least one" = 1 − P(none).**
- **Odds in favour** = P : (1 − P).
- Two dice: 36 outcomes. Sums 2 to 12 have 1, 2, 3, 4, 5, 6, 5, 4, 3, 2, 1 ways. A pack of cards: 52 cards, 4 suits of 13, 12 face cards, 4 kings.

Two fair dice are thrown. Find the probability that the sum is 9 or more, and that the sum is prime.

5/18; 5/12

Sum ≥ 9: ways 4 + 3 + 2 + 1 = 10 of 36. Prime sums 2, 3, 5, 7, 11: 1 + 2 + 4 + 6 + 2 = 15 of 36.

A bag contains 4 red and 6 blue balls. Two balls are drawn without replacement. Find the probability that they are of the same colour.

7/15

(4C2 + 6C2)/10C2 = (6 + 15)/45 = 21/45 = 7/15.

A fair coin is tossed 6 times. Probability of at least 2 heads?

57/64

1 − P(0 or 1 head) = 1 − (1 + 6)/64 = 57/64.

From a pack of 52 cards, one card is drawn. Given that it is a face card, what is the probability it is a king?

1/3

Face cards: 12. Kings among them: 4. P = 4/12 = 1/3.

Two cards are drawn without replacement from a pack of 52. Probability that both are hearts?

1/17

13C2/52C2 = 78/1326 = 1/17.

Machine A makes 30% of the items and machine B makes 70%. A makes 5% defective items, B makes 2% defective items. An item is picked and found defective. What is the probability it came from A?

15/29

P(A and defective) = 0.3 × 0.05 = 0.015. P(B and defective) = 0.7 × 0.02 = 0.014. P(A | defective) = 0.015/0.029 = 15/29.

Find the expected value of the sum when two fair dice are thrown.

7

E(one die) = 3.5, so E(sum) = 7.

Three friends A, B, C try to solve a problem with probabilities of solving 1/2, 1/3, 1/4 independently. What is the probability that the problem is solved?

3/4

P(none solves) = (1/2)(2/3)(3/4) = 1/4. So P(solved) = 3/4.

How these questions appear

Pattern	Wording	Method
Dice and coins	“Sum is 8”, “at least one six”	List outcomes or use complement.
Cards	“Both are kings”, “one red, one face”	Combinations, or sequential multiplication.
Balls from a bag	“Without replacement”, “with replacement”	Combinations vs repeated fractions.
Conditional	“Given that ...”	Restrict the sample space.
Bayes	Factories, tests, defective items	Weighted fractions.

Binomial	“Exactly 3 heads in 5 tosses”	nCk p^k q^{n-k}.
Meeting/geometric	Birthdays, same-sex pairs	Complement trick.

Shortcuts and tricks

- For “at least one” always use 1 − P(none).
- Without replacement, probabilities of later draws change; with replacement they do not.
- For conditional probability redraw the sample space after the condition.
- Give the answer as a reduced fraction unless the question asks for a decimal or percentage.

Common mistakes

- Adding probabilities of events that are not mutually exclusive.
- Forgetting that without replacement the denominator reduces.
- Treating dependent events as independent.

Practice set: Probability practice

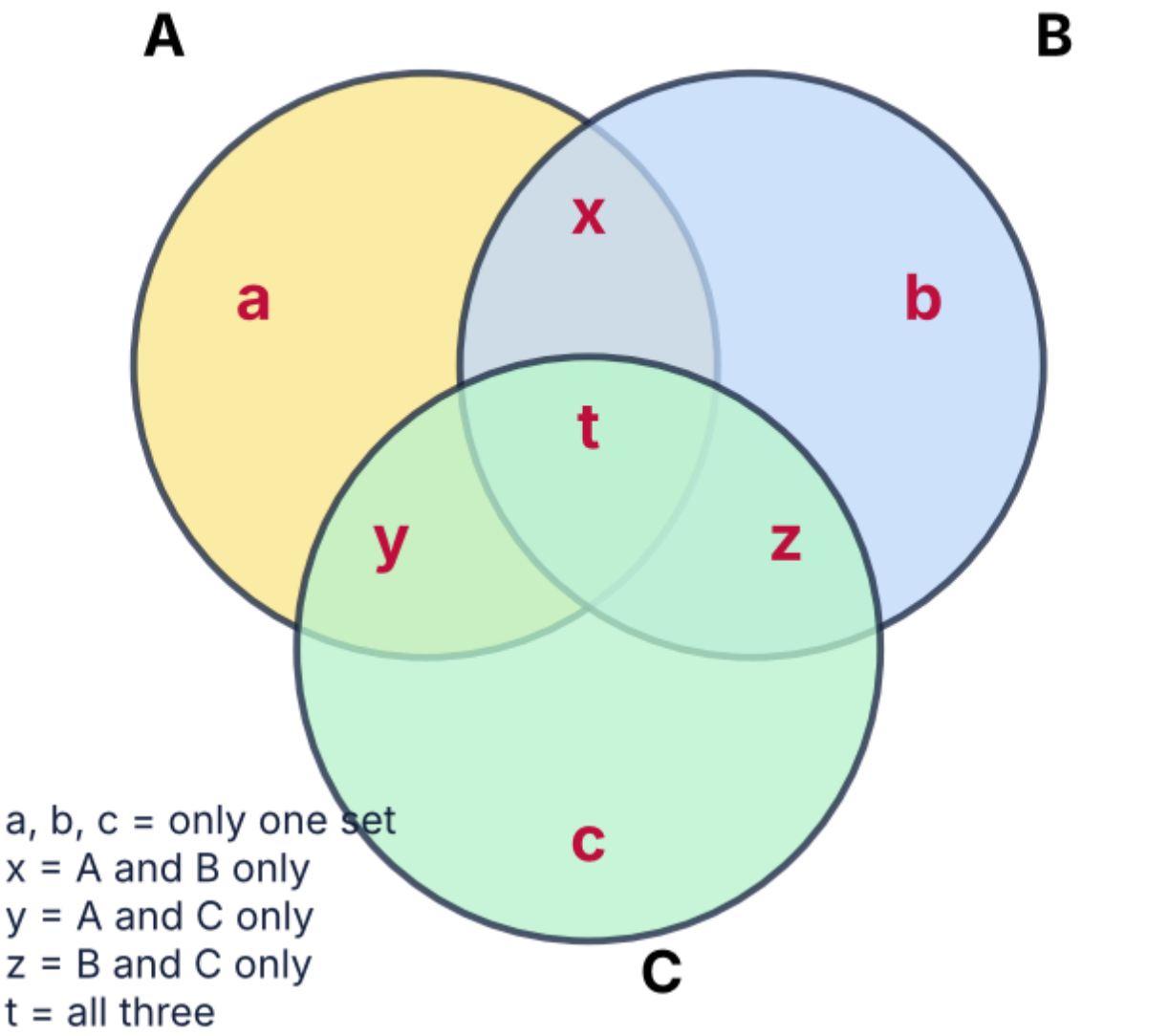
1. A die is thrown once. Probability of an even number greater than 2? **Basic**
2. Two coins are tossed. Probability of exactly one head? **Basic**
3. Probability that a card drawn from a pack is a red king? **Basic**
4. A bag has 5 red and 3 green balls. Two drawn together. Probability both red? **Medium**
5. Three dice are thrown. Probability that the sum is 6? **Medium**
6. A coin is tossed 5 times. Probability of exactly 3 heads? **Medium**
7. Probability that in a group of 3 people at least two share a birth day of the week? **Advanced**
8. A and B take turns tossing a coin; A starts; first head wins. Probability A wins? **Advanced**
9. Two dice are thrown. Given that the sum is 8, probability that both numbers are even? **Advanced**
10. Probability that a number selected from 1 to 100 is divisible by 4 or 6? **Advanced**

08 Set Theory and Venn Diagrams

ADVANCED QUANT · SETS · IMPORTANT

Set questions give you survey data (how many like tea, coffee, both...) and ask for a region of the Venn diagram. The method is always the same: **draw the diagram, fill the innermost region first, then work outwards**. Advanced questions add “at least”, “exactly” and “minimum/maximum” overlaps.

Three-set Venn diagram: seven regions + outside



Fill t (all three) first, then x, y, z, then the single regions.

Formulas

- **$n(A \cup B) = n(A) + n(B) - n(A \cap B)$.**
- **$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(A \cap C) + n(A \cap B \cap C)$.**
- Only A = $n(A) - n(A \cap B) - n(A \cap C) + n(A \cap B \cap C)$.
- **Exactly two sets** = Σ pairs − 3 × (all three). **Exactly one set** = Σ singles − 2 × Σ pairs + 3 × (all three).
- **At least two sets** = Σ pairs − 2 × (all three).
- **Neither** = Total − $n(A \cup B \cup C)$.
- **Minimum overlap** of A and B in a universe U: $\max(0, n(A) + n(B) - U)$. **Maximum overlap**: $\min(n(A), n(B))$.
- For percentages, take the total as 100.

In a group, 60 like tea, 50 like coffee and 40 like juice. 20 like tea and coffee, 15 coffee and juice, 18 tea and juice, and 8 like all three. Find (a) how many like at least one drink, (b) only tea, (c) exactly two drinks, (d) exactly one drink.

105; 30; 29; 68

(a) 60+50+40 − 20 − 15 − 18 + 8 = 105. (b) Only tea = 60 − 20 − 18 + 8 = 30. (c) Exactly two = 20+15+18 − 3×8 = 29. (d) Exactly one = 150 − 2×53 + 3×8 = 68.

In a class of 80 students, 55 like cricket and 45 like football. What is the minimum and the maximum number who like both?

20; 45

Minimum overlap = 55 + 45 − 80 = 20 (everyone likes at least one). Maximum overlap = the smaller set = 45.

In a survey of 200 people, 120 read The Hindu, 90 read the Times and 30 read neither. How many read both?

40

Readers of at least one = 200 − 30 = 170. Both = 120 + 90 − 170 = 40.

In a town 60% read paper A, 50% read paper B and 40% read paper C. 30% read A and B, 20% read B and C, 25% read A and C, and 10% read all three. What percentage read none?

15

Union = 60 + 50 + 40 − 30 − 20 − 25 + 10 = 85%. None = 100 − 85 = 15%.

Counting numbers divisible by given numbers (a favourite)

How many integers from 1 to 500 are divisible by 3 or 5 or 7?

271

Use inclusion-exclusion with $\lfloor 500/k \rfloor$: 166 + 100 + 71 − 33 − 23 − 14 + 4 = 271.

How many integers from 1 to 500 are divisible by none of 3, 5 and 7?

229

500 − 271 = 229.

How these questions appear

Pattern	Wording	Method
Survey, 2 sets	"x read A, y read B, z read neither"	Union = total − neither.
Survey, 3 sets	Given pairs and all three	Seven-region diagram.
Exactly / at least	"Exactly two of the three"	Pair sums − 3 × triple.
Min / max overlap	"Minimum number who like both"	Overlap bounds.
Divisibility counting	"Numbers up to 1000 divisible by 4 or 6"	Floors and inclusion–exclusion.
Percentage sets	"60% like A ..."	Use total 100.

Shortcuts and tricks

- Always compute the region of all three first.
- A missing region = the given total minus all other regions. Check that your seven numbers add to the total.
- For "either ... or" divisibility count, subtract the multiples of the LCM.

Common mistakes

- Subtracting $n(A \cap B)$ from $n(A)$ to get "only A" when C also overlaps (you must add back the triple).
- Reading "like tea and coffee" as "only tea and coffee".
- Mixing percentages with actual numbers.

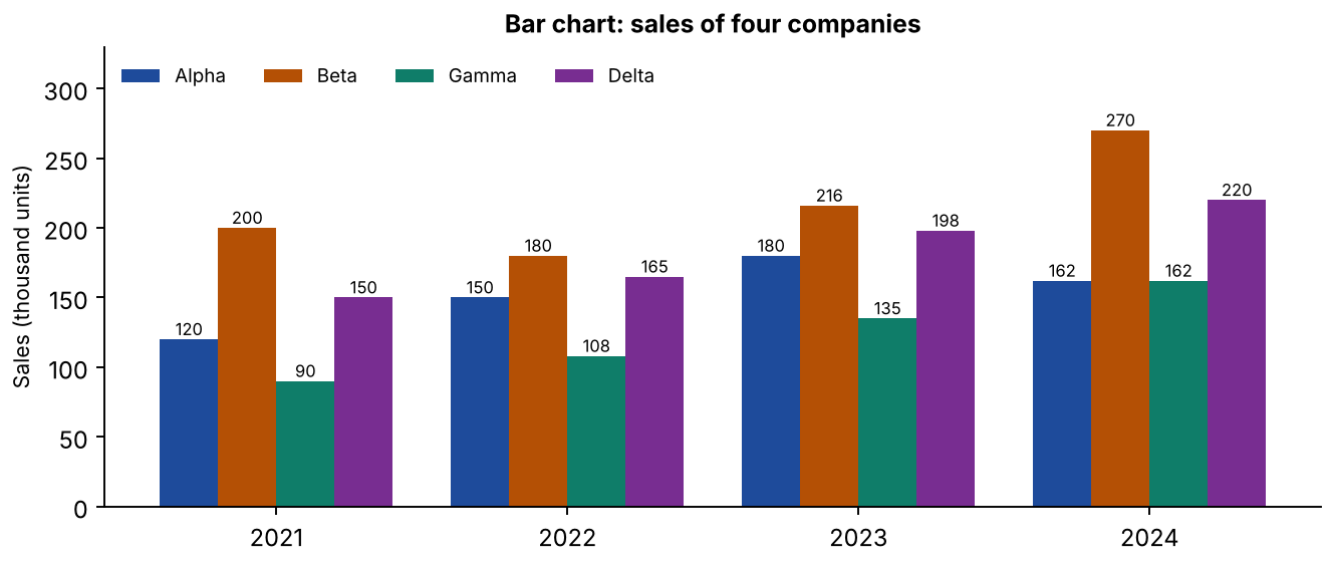
Practice set: Set theory practice

1. $n(A) = 30$, $n(B) = 25$, $n(A \cap B) = 10$. Find $n(A \cup B)$. — Basic
2. In a class of 50, 30 take Maths, 25 take Physics and 8 take neither. How many take both? — Basic
3. How many integers from 1 to 100 are divisible by 4 or 6? — Medium
4. Of 100 people, 70 like tea and 60 like coffee. Minimum number who like both? — Medium
5. Maximum number who like both, same data? — Medium
6. With A=40, B=35, C=30, AB=12, BC=10, AC=9, ABC=5, find exactly two sets. — Medium
7. Same data: how many like only C? — Medium
8. How many integers from 1 to 1000 are divisible by 2, 3 or 5? — Advanced
9. How many integers from 1 to 1000 are divisible by none of 2, 3, 5? — Advanced
10. In a group of 100, 45 know French, 40 know German, 30 know Spanish, 12 French and German, 10 German and Spanish, 8 French and Spanish, 4 all three. How many know none? — Advanced

09 Data Interpretation

ADVANCED QUANT · DATA · MUST-DO

A Data Interpretation (DI) set is a table or chart followed by several questions. The numbers are not hard; the **reading and the percentage arithmetic under time pressure** are. The method is: read the title and units, locate the numbers asked, and use **approximation** when the options are far apart.



Bar chart. Check the unit (thousand units) before you calculate.

Formulas used in DI

- **Percentage change = (new − old)/old × 100.** Always divide by the OLD value.
- **Percentage of total = part/total × 100.**
- **Ratio:** compare values by dividing; ratios of percentages equal ratios of values when the total is the same.
- **Average = sum/number of items.** Weighted average for groups of different sizes.
- **Percentage points** vs **percent**: a rise from 12% to 18% is 6 percentage points but 50% in relative terms.
- **CAGR (growth per year)** for 2 years: $(1 + g)^2$ = final/initial.
- Pie chart: a slice of x% has an angle of 3.6 × x degrees.
- Table with missing cells: use row and column totals to find them.

From the bar chart, find the percentage increase in the sales of Beta from 2023 to 2024.

25%

(270 − 216)/216 × 100 = 54/216 × 100 = 25.00%.

What percentage of the total sales of all four companies in 2023 was the sales of Gamma?

18.52%

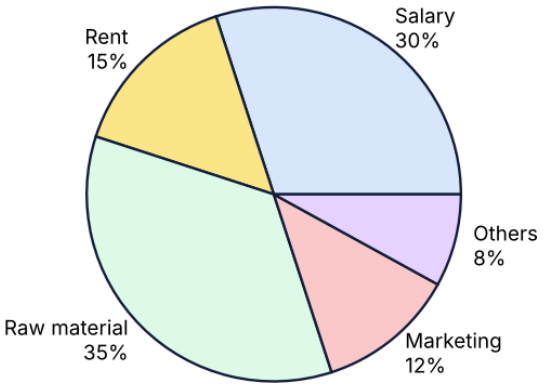
Total in 2023 = 180 + 216 + 135 + 198 = 729. Gamma = 135/729 × 100 = 18.52%.

What is the average yearly sales of Alpha, and which company showed the largest absolute increase from 2021 to 2024?

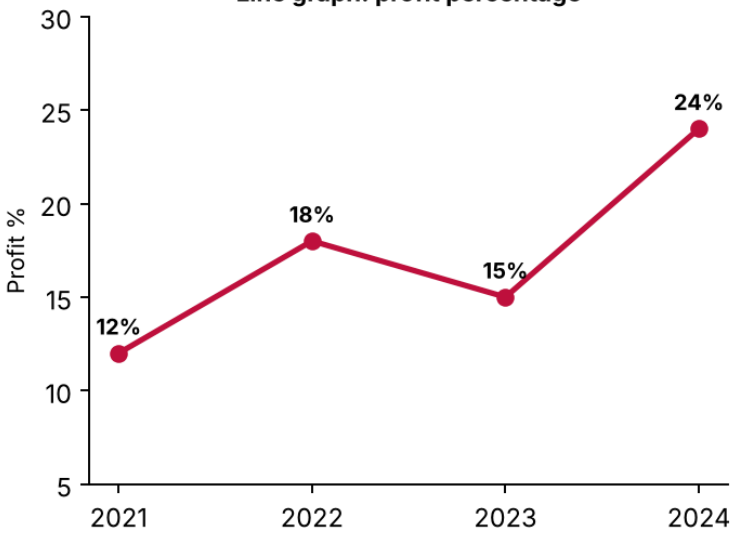
153.0; Gamma

Alpha: (120 + 150 + 180 + 162)/4 = 153.0. Increases 2021→2024: Alpha 42, Beta 70, Gamma 72, Delta 70. Largest: Gamma.

Pie chart: expenses of a firm (total ₹2,40,000)



Line graph: profit percentage



Pie chart and line graph. Pie slices are percentages of one total; the angle of a slice is 3.6° per percent.

In the pie chart, what amount (in ₹) is spent on salary, and what is the central angle of the Raw material slice?

₹72,000; 126.0°

Salary = 30% of ₹2,40,000 = ₹72,000. Raw material = 35% × 3.6° = 126.0°.

From the line graph, by what percentage points did the profit rise from 2021 to 2022, and by what percent in relative terms?

6 points; 50%

18 − 12 = 6 percentage points. Relative rise = 6/12 × 100 = 50%.

If the 2024 sales of Delta were 12% more than the 2023 sales of Delta, how much is the 2024 figure compared with the chart?

221.76 vs 220

198 × 1.12 = 221.76, which is higher than the 220 shown. This shows how a “what if” question is checked against the chart.

Table DI: missing values

A table shows the number of items sold in four months by three stores. Row totals are 150, 170, 160. Row 1: 32, 45, ?, 28. Row 2: 40, ?, 36, 30. Row 3: ?, 52, 41, 35. Find the three missing numbers.

45, 64, 32

Row 1: 150 − (32 + 45 + 28) = 45. Row 2: 170 − (40 + 36 + 30) = 64. Row 3: 160 − (52 + 41 + 35) = 32.

Speed techniques for DI

Situation	Trick
Percentage change	Cancel common factors first (216 → 270: both divisible by 54 → 4 → 5 → 25%).
Compare two ratios	Cross-multiply instead of dividing.
Which is largest?	Estimate; do not compute exactly unless close.
Percent of a total	Use fractions: 12.5% = 1/8, 16.67% = 1/6, 37.5% = 3/8.
Pie chart with values	Convert angles: 90° = 25%, 120° = 33.33%.
Average of many values	Pick a base and add small differences.

Shortcuts and tricks

- Read the units and the legend first. A thousand-versus-lakh slip costs the whole set.
- Underline “increase”, “decrease”, “compared with”, “as a percentage of”. Each fixes the base of the percentage.
- If two options are within 2%, calculate exactly; otherwise estimate.
- For “which year had the highest …”, compare only the likely candidates.

Common mistakes

- Dividing by the new value instead of the old one in a percentage change.
- Confusing percentage points with percent.
- Using the wrong year’s total.

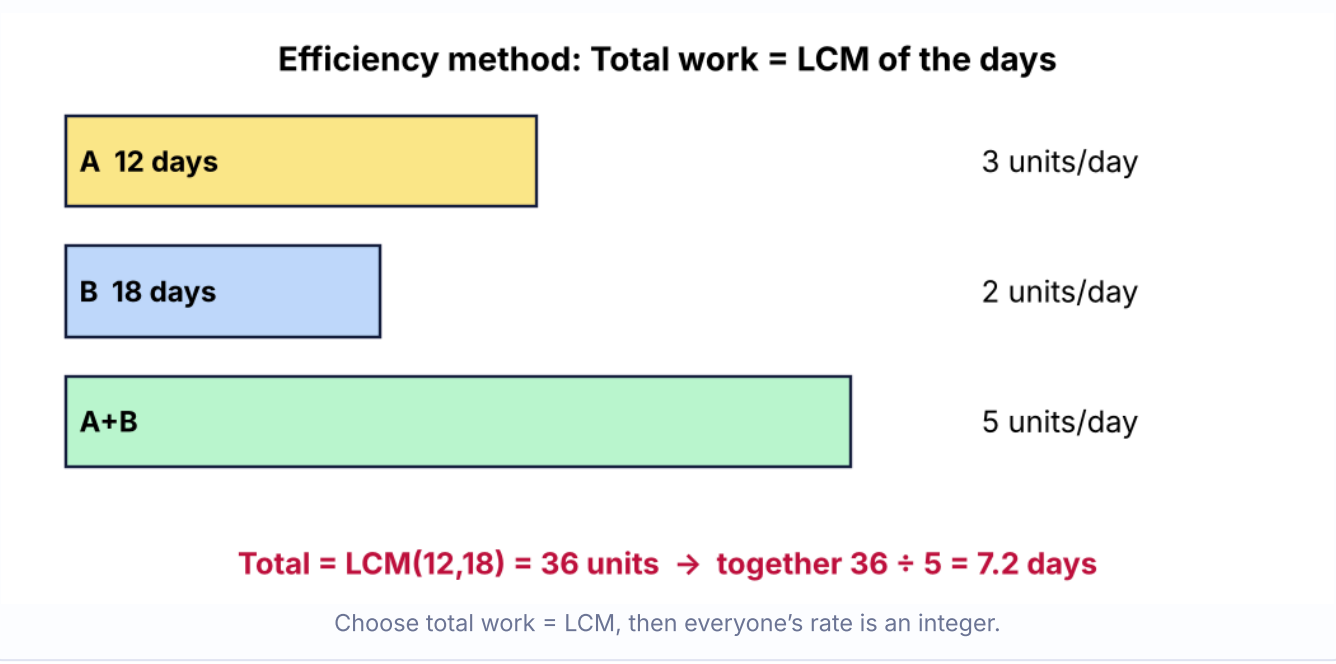
Practice set: DI practice (use the charts in this chapter)

1. Sales of Alpha in 2022 as a percentage of its sales in 2021? — Basic
2. Total sales of all companies in 2021? — Basic
3. Which company had the highest sales in 2024? — Basic
4. Percentage decrease in Alpha’s sales from 2023 to 2024? — Medium
5. Ratio of Beta’s 2022 sales to Gamma’s 2022 sales? — Medium
6. Average sales of Gamma over the four years? — Medium
7. Rent expense in ₹ (pie chart)? — Medium
8. Central angle for Marketing? — Medium
9. In which year was the combined sales of Alpha and Delta the highest? — Advanced
10. CAGR-style: if Beta grows at the same yearly rate as 2022→2024 (two years), what is the single yearly growth factor g where 1+g squared = 270/180? — Advanced

ADVANCED QUANT · TIME AND WORK · HIGH PRIORITY

10 Time and Work, Pipes and Cisterns

Every Time-and-Work question is one idea: **work = rate × time**. The trick that makes it easy is the **efficiency (units) method**: pick the total work as the LCM of the given days, so every rate becomes a whole number. Pipes and cisterns, wages, and alternate-day questions are all the same idea.



Choose total work = LCM, then everyone's rate is an integer.

Core formulas

- **Work = Rate × Time.** If A finishes in a days, A's rate = 1/a per day (or LCM/a units per day).
- **Together:** 1/a + 1/b = (a+b)/(ab), so time = **ab/(a+b)** for two people.
- If A and B together take t days and A alone takes a days: B alone = **at/(a−t)**.
- **Efficiency ∝ 1/days.** If A is k times as efficient as B, A takes 1/k of B's time.
- **Wages are divided in the ratio of work done** (efficiency × days), not in the ratio of days alone.
- $M_1D_1H_1/W_1 = M_2D_2H_2/W_2$ (men × days × hours per day ÷ work).
- **Pipes:** inlet adds, outlet / leak subtracts. Net rate = Σ inlets − Σ outlets.
- **Alternate-day work:** compute work done in one 2-day cycle, count full cycles, then handle the remainder.

A can do a piece of work in 12 days and B in 18 days. In how many days can they do it together?

36/5 = 7.2 days

Total = LCM(12,18) = 36. A: 3/day, B: 2/day. Together 5/day. Time = 36/5 = 36/5 days = 7.2 days.

A and B together finish a job in 8 days. A alone takes 24 days. How long will B alone take?

12

Use B = at/(a−t) = 24×8/(24−8) = 12 days. Units check: total 24; together 3/day; A 1/day; B 2/day → 12 days.

A, B and C can do a job in 10, 15 and 30 days respectively. A and B start; after 3 days A leaves and C joins. How many more days are needed to finish?

5

Total = LCM = 30. Rates: A 3, B 2, C 1. First 3 days: 3 × 5 = 15. Remaining 15. B + C = 3/day → 5 days.

A can do a work in 12 days and B in 18 days. They work on alternate days, A starting first. In how many days is the work finished?

15

Total = 36. A = 3/day, B = 2/day. One 2-day cycle = 5 units. 6 cycles (12 days) = 30. Day 13 (A): 33. Day 14 (B): 35. Day 15 (A): 38 ≥ 36. So it finishes on day 15.

15 men can complete a job in 24 days working 8 hours a day. How many days will 20 men take working 6 hours a day?

24

Man-hours = 15×24×8 = 2880. New rate per day = 20×6 = 120. Days = 2880/120 = 24.

A can do a work in 15 days; B is 50% more efficient than A. Find B's time.

10.0

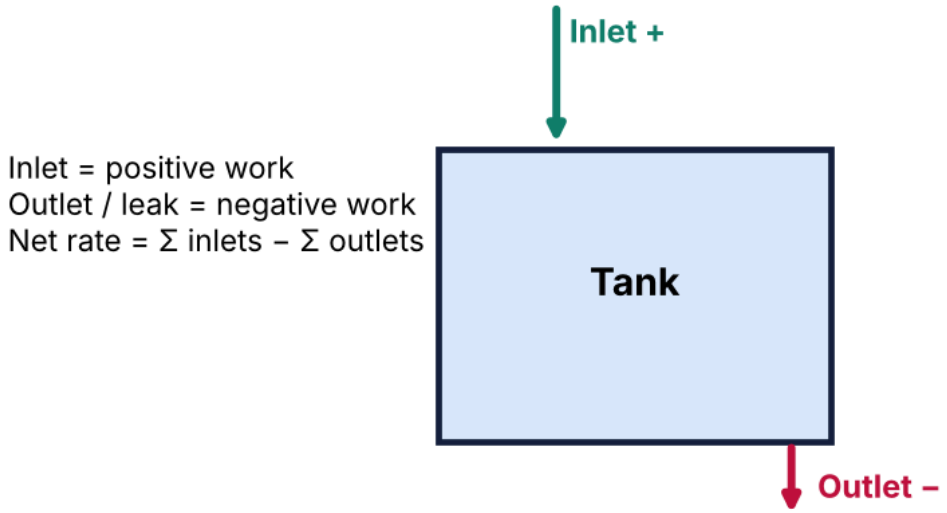
B is 1.5× as efficient, so B takes 15/1.5 = 10 days.

A and B together earn ₹1,800 for a job. A works 6 days at efficiency 3, B works 9 days at efficiency 2. What is A's share?

900

Work done: A = 6×3 = 18, B = 9×2 = 18. Equal, so A gets ₹900.

Pipes and cisterns



Inlet adds, outlet subtracts.

A pipe fills a tank in 6 hours and a leak empties it in 10 hours. How long will the tank take to fill with the leak open?

15

Total = LCM(6,10) = 30. Inlet 5/hr, leak 3/hr, net 2/hr. Time = 30/2 = 15 hours.

Pipes A and B fill a tank in 20 and 30 minutes; C empties it in 40 minutes. All open together: time to fill?

120/7

Total = LCM(20,30,40) = 120. A 6, B 4, C −3 → net 7/min. Time = 120/7 ≈ 17.14 minutes.

How these questions appear

Pattern	Wording	Method
Joint work	"A in 10, B in 15, together?"	ab/(a+b) or LCM units.
Leaving / joining	"A leaves after 3 days"	Work done so far, then the remainder.
Alternate days	"work on alternate days"	One cycle total; count cycles.
Efficiency	"20% more efficient"	Efficiency ∝ 1/days.
Wages	"Share the ₹ in proportion"	Split by work done.
Men-days	"15 men, 24 days, 8 hr"	M·D·H/W constant.
Pipes	"leak empties it in ..."	Net rate, outlet negative.

Shortcuts and tricks

- Total work = LCM of the given days: always, no exceptions.
- "k times as efficient" means ratio k : 1 and time ratio 1 : k.
- For leaving/joining, compute the units done by each phase and subtract from the total.
- In pipe questions with a leak, a tank that "fills in x hours but takes y with the leak" gives leak rate = 1/x − 1/y.

Common mistakes

- Dividing wages by days instead of by efficiency × days.
- Treating a leak as an inlet (sign error).
- In alternate-day work, forgetting the work might finish partway through a day.
- Adding days instead of rates.

Practice set: Time-and-work practice

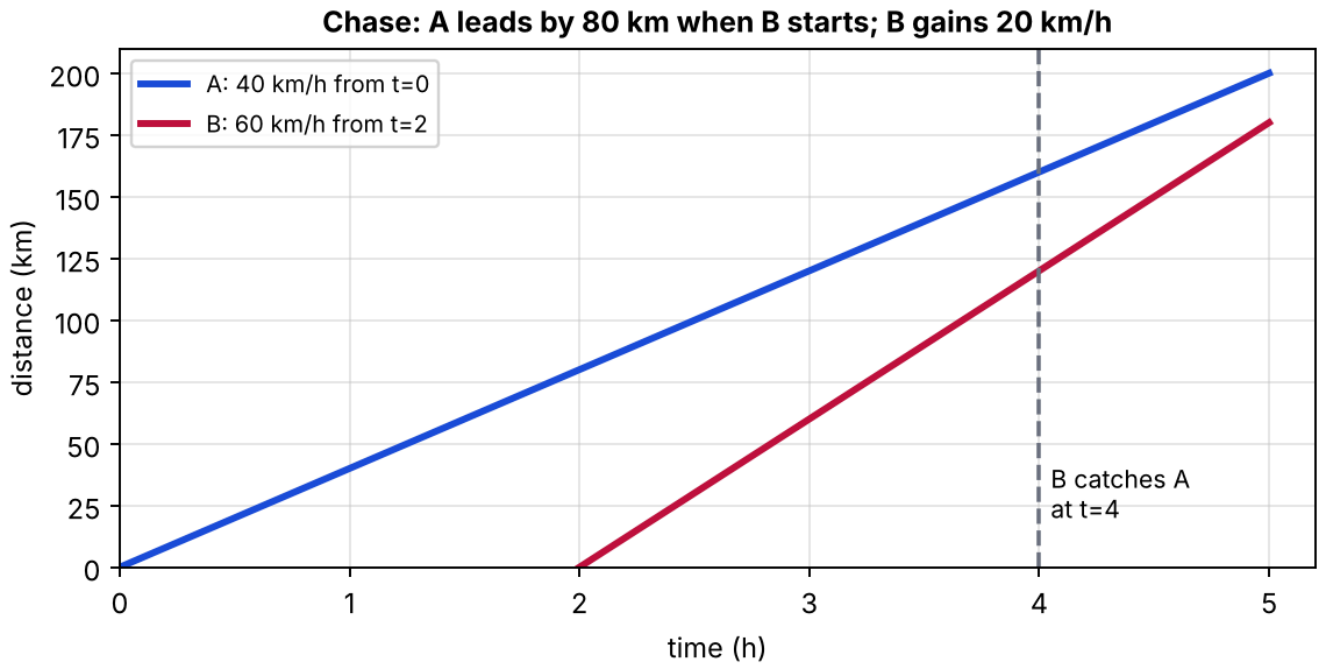
1. A does a job in 20 days, B in 30 days. Together? **Basic**
2. A is twice as fast as B. Together they take 12 days. How many days would A alone take? **Basic**
3. A pipe fills a tank in 8 hours. After 2 hours another pipe, which fills in 12 hours, is also opened. How long from the start to fill? **Medium**
4. A and B can do a job in 12 and 16 days. They work for 4 days; then B leaves. How long does A need to finish alone? **Medium**
5. 12 men finish a job in 20 days. After 5 days, 6 more men join. In how many more days will it finish? **Medium**

6.	A can do a work in 10 days, B in 12 days and C in 15 days. They work together; A leaves 2 days before completion. Total days?	Advanced
7.	Two pipes fill a tank in 15 and 20 hours; a third empties it in 30. If all three are opened together, how long to fill?	Advanced
8.	A can do a piece of work in 24 days. With B's help it takes 8 days. In how many days can B alone do it?	Advanced
9.	A and B alternately work, A first. A alone takes 10 days, B alone takes 15. When is the job done?	Advanced
10.	Time ratio of A to B is 3:4. A and B together finish in 12 days. How long would B alone take?	Advanced

ADVANCED QUANT · TSD · HIGH PRIORITY

11 Time, Speed and Distance (Advanced)

Speed questions look different but reduce to **Distance = Speed × Time** and three tools: **relative speed** (chase and meeting), **average speed** (harmonic, never arithmetic), and **ratio of speeds ↔ inverse ratio of times**. Trains, boats and races are applications.



A distance–time graph: where the lines cross is where they meet.

Formulas

- **km/h → m/s:** multiply by 5/18. **m/s → km/h:** multiply by 18/5.
- **Average speed** for equal distances at speeds a and b = **2ab/(a+b)**. For equal times it is (a+b)/2.
- **Relative speed:** same direction = difference; opposite direction = sum.
- **Train crossing a pole/person:** time = train length ÷ speed. **Crossing a platform/bridge:** time = (train + platform) ÷ speed.
- **Two trains:** same direction, time = $(L_1+L_2)/(v_1-v_2)$; opposite, $(L_1+L_2)/(v_1+v_2)$.
- **Boats:** downstream = u + v, upstream = u − v. Still water speed u = (down + up)/2; stream v = (down − up)/2.
- **Equal distance:** speeds ratio a : b ⇒ times ratio b : a.
- **Meeting two persons** from A and B towards each other: distance ÷ (sum of speeds).
- **Circular track:** meet first time at the opposite direction after $L/(v_1+v_2)$, same direction $L/(v_1-v_2)$.

A car goes from P to Q at 40 km/h and returns at 60 km/h. Find the average speed for the whole trip.

48

Equal distances, so harmonic mean: 2×40×60/(40+60) = 48 km/h.

A 150 m train moving at 54 km/h crosses a 250 m platform. How long does it take?

80/3

54 km/h = 54×5/18 = 15 m/s. Distance = 150 + 250 = 400 m. Time = 400/15 = 80/3 ≈ 26.67 s.

Two trains of length 120 m and 180 m run in opposite directions at 60 and 40 km/h. In how many seconds do they cross?

54/5

Relative speed = 100 km/h = 100×5/18 m/s = 250/9 m/s. Distance = 300 m. Time = 300 × 9/250 = 10.8 s.

Same two trains in the same direction: crossing time?

54

Relative speed = 20 km/h = 50/9 m/s. Time = 300 × 9/50 = 54 s.

A boat goes 24 km downstream in 2 hours and 24 km upstream in 3 hours. Find still-water speed and stream speed.

10; 2

Downstream 12 km/h, upstream 8 km/h. Still water = (12+8)/2 = 10; stream = (12−8)/2 = 2 km/h.

A starts at 40 km/h. Two hours later B starts from the same place in the same direction at 60 km/h. When does B catch A (after B starts)?

4

Head start = 40×2 = 80 km. Closing speed = 20 km/h. Time = 80/20 = 4 hours after B starts.

A man walks at 5 km/h and reaches 10 minutes late; at 6 km/h he reaches 5 minutes early. Find the distance.

15/2

d/5 − d/6 = 15 min = 1/4 h → d/30 = 1/4 → d = 7.5 km.

A can run a 1 km race 100 m ahead of B (A wins by 100 m). If A takes 80 s, how long does B take for the full 1 km?

800/9

When A runs 1000 m, B runs 900 m. B's speed = 900/80 m/s. B needs 1000 ÷ (900/80) = 88.89 s.

Two persons walk around a circular track of length 400 m at 4 m/s and 6 m/s. When do they meet first, in the same direction and in opposite directions?

200 s; 40 s

Same direction: 400/(6−4) = 200 s. Opposite: 400/(6+4) = 40 s.

Pattern	Wording	Method
Average speed	"goes at x and returns at y"	Harmonic mean 2xy/(x+y).

Train/platform	“crosses a 250 m platform”	Add the lengths.
Two trains	“in opposite directions”	Add speeds, add lengths.
Boats	“down in 2 h, up in 3 h”	Down and up speeds → u, v.
Chase	“starts 2 h later”	Head start ÷ closing speed.
Late/early	“5 km/h late, 6 km/h early”	Set up difference of times.
Races	“A beats B by 100 m”	Distance ratio = speed ratio.
Circular	“meet for the first time”	Length ÷ relative speed.

Shortcuts and tricks

- Convert units first: m/s vs km/h is the number-one trap.
- If the distance is fixed, speed × time is constant: use ratios, no equations.
- For “late and early” questions, subtract the two times and equate to the total gap.
- Speeds 5 : 6 means times 6 : 5 — no algebra needed.

Common mistakes

- Using the arithmetic mean for average speed with equal distances.
- Forgetting the train’s own length when crossing a platform.
- Using the wrong sign for boat upstream and downstream.
- Mixing minutes with hours.

Practice set: Speed practice

1. Convert 72 km/h to m/s. — Basic

2. A train 200 m long crosses a pole in 10 s. Speed in km/h? — Basic

3. Average speed if a man goes at 30 km/h and returns at 20 km/h? — Medium

4. A boat goes at 15 km/h in still water and the stream is 3 km/h. Time to go 36 km downstream and come back? — Medium

5. Two trains 150 m and 100 m run in the same direction at 72 and 54 km/h. Time to cross? — Medium

6. A walks at 4 km/h; B starts 3 hours later at 10 km/h. How far from the start does B catch A? — Medium

7. A train crosses a man in 9 s and a 180 m platform in 18 s. Find its length. — Advanced

8. Two cyclists start from points 120 km apart and move towards each other at 14 and 16 km/h. When do they meet? — Advanced

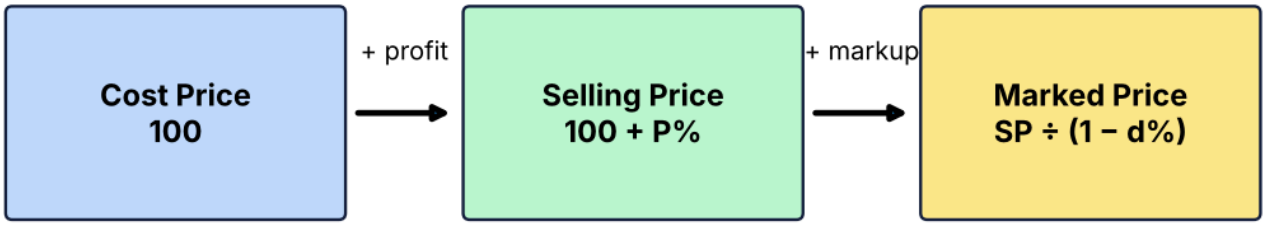
9. A man covers half the distance at 20 km/h and the rest at 30 km/h. His average speed? — Advanced

10. In a 200 m race A beats B by 20 m or 4 s. Find A’s time. — Advanced

ADVANCED QUANT · PERCENT AND MIXTURES · HIGH PRIORITY

12 Percentages, Profit and Loss, Mixtures, Partnership

This chapter bundles the topics that are all **ratio thinking in disguise**: percentages, profit and loss, discounts, alligation (mixtures) and partnership. Convert every percentage to a multiplier (20% up = ×1.2, 20% down = ×0.8) and most questions become one multiplication.



Profit / loss % is always on CP. Discount % is always on MP.

Know which base each percentage is applied on.

Formulas

- **Profit% = (SP − CP)/CP × 100**; **Loss% = (CP − SP)/CP × 100**. SP = CP × (1 ± p/100).
- **Discount% = (MP − SP)/MP × 100**, SP = MP × (1 − d/100).
- **Successive changes** a% and b%: net = a + b + ab/100 (use signs).
- **Two successive discounts** d_1 and d_2: single discount = $d_1 + d_2 - d_1d_2/100$.
- **Equal SP, one gain x% and one loss x%** ⇒ always a loss of (x/10)%.
- **Population / depreciation:** $P \times (1 \pm r/100)^n$.
- **Alligation:** cheaper : dearer = (d − m) : (m − c).
- **Replacement of mixture:** after n replacements of a fraction x of a container, remaining = $V(1 - x)^n$.
- **Partnership:** profit share ∝ investment × time.
- If x% of A is y% of B then A : B = y : x.

A shopkeeper marks an article 25% above cost and gives 20% discount. Find the profit or loss percentage.

0

Let CP = 100. MP = 125. SP = 125 × 0.8 = 100. Profit = 0%.

Price rises by 20% then falls by 20%. Net change?

-4

1.2 × 0.8 = 0.96, a net 4% decrease. Formula: 20 − 20 − 400/100 = −4.

A man sells two articles at ₹1,000 each; one at 25% profit and one at 25% loss. Overall result?

2133.33 cost; loss -133.33

CP_1 = 1000/1.25 = 800, CP_2 = 1000/0.75 = 1333.33. Total CP = 2133.33, SP = 2000. Loss = 133.33, i.e. (25/10)% = 6.25% of the total cost.

Two successive discounts of 20% and 10% are equal to a single discount of?

28

20 + 10 − 20×10/100 = 28%.

In what ratio must rice at ₹40/kg be mixed with rice at ₹60/kg to get a mixture worth ₹48/kg?

12 : 8

Cheaper : dearer = (60−48) : (48−40) = 12 : 8 = 3 : 2.

A vessel has 80 litres of milk. 20 litres are removed and replaced with water, and the process is repeated once. How much milk is left?

45

Remaining = 80 × $(1 - 20/80)^2$ = 80 × $(3/4)^2$ = 45 litres.

A, B and C invest ₹20,000, ₹30,000 and ₹50,000. B joins after 4 months, C after 6 months of a year. In what ratio is a profit divided?

240000 : 240000 : 300000

A: 20000×12 = 240000; B: 30000×8 = 240000; C: 50000×6 = 300000. Ratio 8 : 8 : 10 = 4 : 4 : 5.

If 40% of A = 25% of B, find A : B.

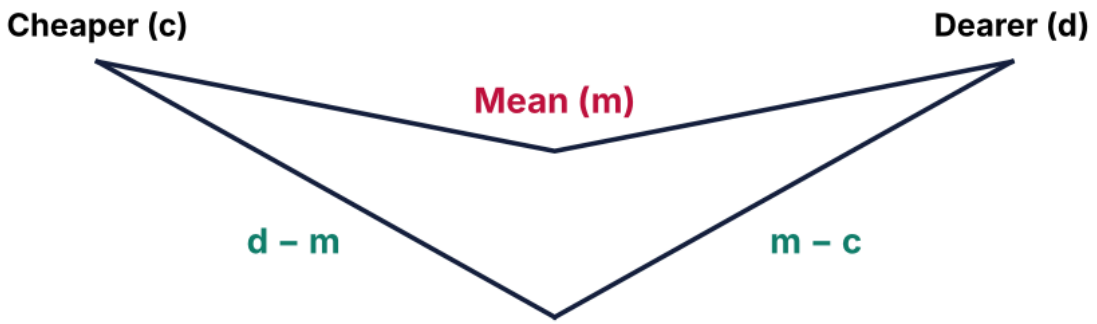
5/8

A : B = 25 : 40 = 5 : 8.

A population of 50,000 grows 10% a year. What is it after 2 years?

60500

50000 × 1.1^2 = 60,500.



Quantity of cheaper : Quantity of dearer = (d – m) : (m – c)

The alligation cross: write cheaper and dearer, subtract diagonally.

Pattern	Wording	Method
Chain of changes	“up 20% then down 20%”	Multiply the factors.
Mark-up and discount	“marked 25% above, discount 20%”	Take CP = 100.
Two articles	“same SP, gain and loss”	Loss of (x/10)%.
Alligation	“mix to get ₹48/kg”	Cross method.
Replacement	“removes 20 L and replaces”	$V(1 - x)^n$.
Partnership	“joins after 4 months”	Money × months.
Percent of percent	“x% of A is y% of B”	Ratio y : x.

Shortcuts and tricks

- Assume the base as 100 (or 1000) to avoid fractions.
- Percentages up then down never cancel.
- Alligation always needs the three prices: cheap, dear and mean.
- If the discount is on the marked price and profit on cost, choose the base accordingly.

Common mistakes

- Applying profit % on selling price.
- Adding successive percentages directly.
- Mixing the order of the alligation cross (cheap is subtracted from mean in the dear column).
- Ignoring time in partnership when partners join late.

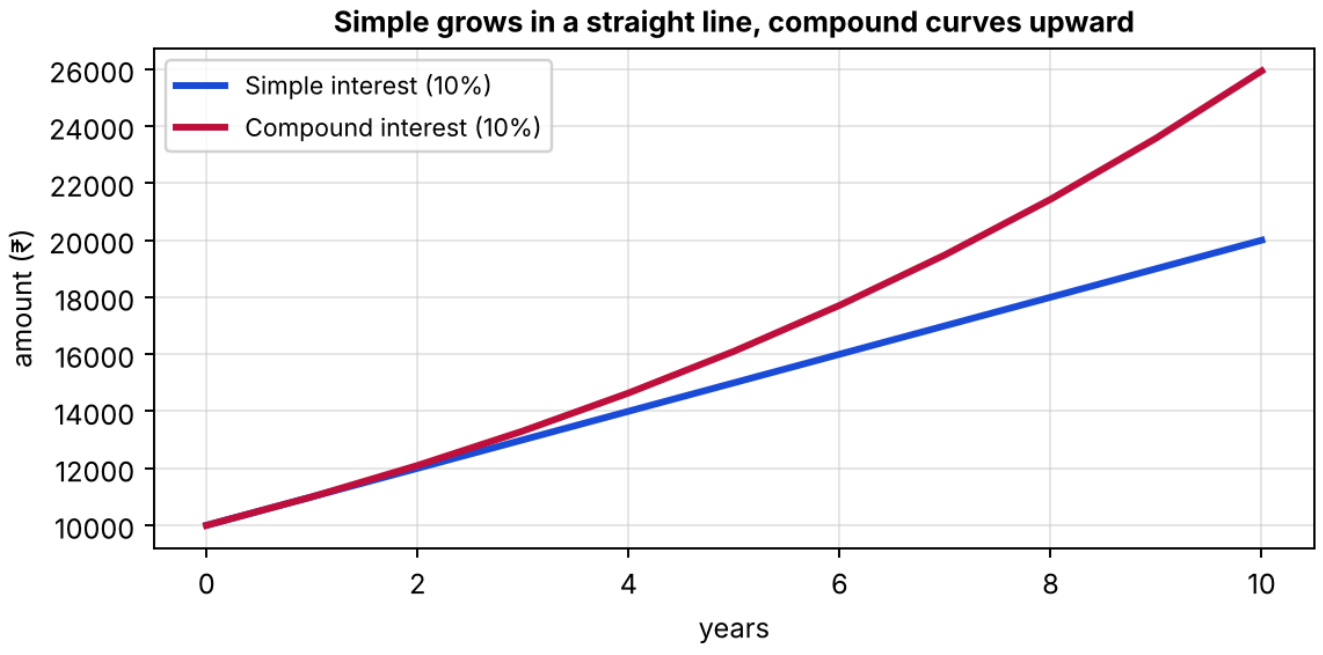
Practice set: Practice set

1. A number is increased by 25% and then decreased by 20%. Net change in %? **Basic**
2. CP = ₹500, SP = ₹600. Profit %? **Basic**
3. Two successive discounts of 10% and 10% equal a single discount of what percent? **Basic**
4. In what ratio must milk at ₹30 be mixed with water (₹0) so the mixture costs ₹24/L? **Medium**
5. A trader marks a product 40% above cost and offers a 30% discount. Profit or loss %? **Medium**

6.	A vessel contains 100 L of milk. 10 L is taken and replaced by water, repeated twice more (3 times total). Milk left?	Medium
7.	A and B invest in the ratio 3 : 5 for 12 and 8 months. Profit ₹7,600; B's share?	Advanced
8.	The price of sugar rises 25%. By what % should consumption fall to keep expenditure the same?	Advanced
9.	A sells at a loss of 10%. If the SP were ₹60 more, there would be a 5% gain. Find CP.	Advanced
10.	The ratio of 2 alloys with 20% and 50% copper to get 35% copper?	Advanced

13 ADVANCED QUANT · INTEREST · MEDIUM PRIORITY
Simple and Compound Interest

Interest has only two formulas worth memorising. The rest is shortcuts: the **difference between CI and SI**, **half-yearly compounding** and **instalments**.



Same rate: compound pulls ahead each year.

Formulas

- **SI = P × R × T / 100.** Amount A = P + SI.
- **CI: A = P(1 + R/100)^n.** CI = A − P.
- **Half-yearly:** rate R/2, time 2n. **Quarterly:** rate R/4, time 4n.
- **CI − SI for 2 years** = P(R/100)^2. For 3 years: P(R/100)^2(3 + R/100).
- **Doubling time (SI):** 100/R years. **Sum becomes k times in t years (SI):** R = 100(k−1)/t.
- **Rule of 72** (approximate, CI): doubling years ≈ 72/R.
- If a sum doubles in n years with CI, it becomes 8 times in 3n years.
- **Same sum at SI for 2 years versus CI:** CI for year 2 = SI for year 1 + interest on interest.

Find the simple interest on ₹8,000 at 6.5% per annum for 3 years.

1560

SI = 8000 × 6.5 × 3 / 100 = ₹1,560.

Find the compound interest on ₹10,000 at 10% p.a. for 2 years.

2100

A = 10000 × 1.1^2 = 12,100. CI = ₹2,100.

Find the difference between CI and SI on ₹5,000 at 8% p.a. for 2 years.

32

$P(R/100)^2$ = 5000 × 0.0064 = ₹32.

₹16,000 at 20% per annum compounded half-yearly for 1 year. Find the amount.

19360

Rate 10% per half-year, 2 periods: 16000 × 1.21 = ₹19,360.

A sum at SI becomes ₹2,400 in 2 years and ₹3,000 in 5 years. Find the sum and the rate.

2000; 10%

SI per year = (3000−2400)/3 = 200. In 2 years SI = 400 so P = 2000. Rate = 200×100/2000 = 10%.

In what time will a sum double at 12.5% SI?

8

Doubling time = 100/R = 100/12.5 = 8 years.

₹25,000 is invested at 8% CI for 3 years. Find the amount.

157464/5

$A = 25000 \times (1.08)^3$ = ₹31,492.80.

Pattern	Wording	Method
Plain SI	"interest on ₹x at r% for t years"	PRT/100.
CI direct	"compounded annually"	$P(1+r)^n - P$.
CI vs SI	"difference between ... for 2 years"	$P(r/100)^2$.
Half-yearly	"compounded half-yearly"	Half rate, double time.
Doubling	"doubles in ... years"	100/R (SI), 72/R (CI).
Two-sum	"same amount lent at SI and CI"	Equate and solve.
Instalments	"paid in 2 equal instalments"	Present value of each.

Shortcuts and tricks

- For 2-year CI use the shortcut $r + r + r^2/100$ as a single percentage.
- CI question with r = 10% gets easy with multiplier 1.1, 1.21, 1.331, 1.4641.
- Convert time to years before using the formulas (months ÷ 12).
- Half-yearly CI is always a little bigger than yearly CI at the same nominal rate.

Common mistakes

- Using R/2 without doubling n for half-yearly compounding.
- Reading "CI" where the question asks for the amount (or the reverse).
- Applying the difference-of-CI-and-SI shortcut to more than 2 years.
- Using months directly in years.

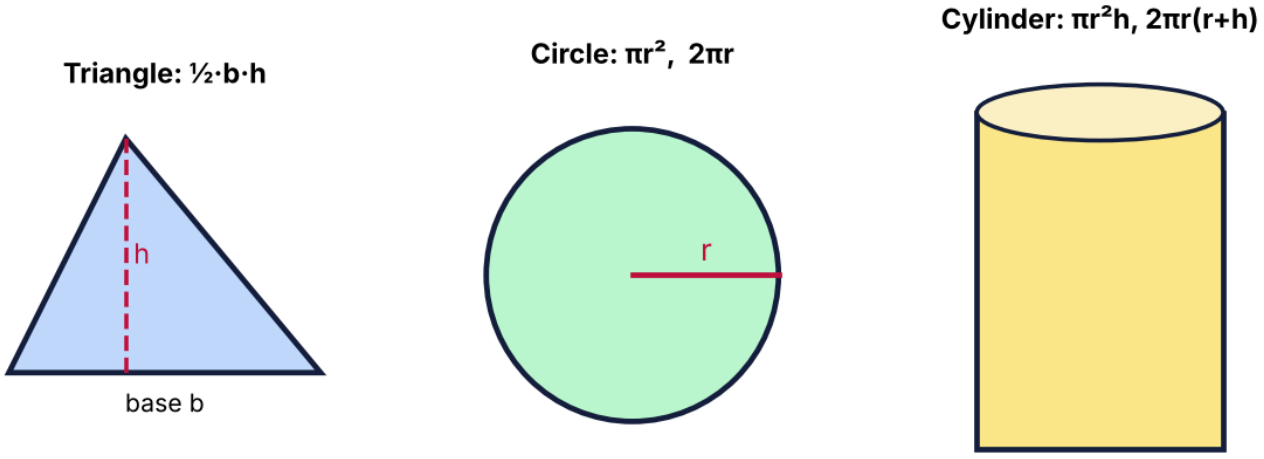
Practice set: Interest practice

1.	SI on ₹6,000 at 5% for 4 years?	Basic
2.	CI on ₹4,000 at 10% for 2 years?	Basic
3.	Amount on ₹12,000 at 5% CI for 2 years?	Basic
4.	CI − SI on ₹10,000 at 5% for 2 years?	Medium
5.	A sum triples in 10 years at SI. Find the rate.	Medium
6.	₹8,000 at 10% p.a. compounded half-yearly for 1 year: CI?	Medium
7.	A sum at CI doubles in 4 years. In how many years will it become 16 times?	Advanced
8.	A person borrows ₹15,000 at 6% SI and lends at 8% SI for 3 years. His gain?	Advanced
9.	The interest earned in year 2 on a sum at 10% CI is ₹1,100. Find the sum.	Advanced
10.	At what rate p.a. will ₹4,000 amount to ₹4,840 in 2 years at CI?	Advanced

14 Mensuration and Geometry

ADVANCED QUANT · GEOMETRY · MEDIUM PRIORITY

Mensuration is a formula-recall topic; geometry questions in NQT are mostly area/perimeter of plane figures, volumes of simple solids, and angle and triangle properties. Memorise the list below and learn the three **ratio rules**: scale factor k ⇒ lengths ×k, areas ×k^2, volumes ×k^3.



The three figures that give most questions.

Plane figures

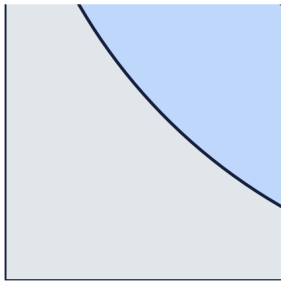
- Rectangle: area lb, perimeter 2(l+b), diagonal √(l^2+b^2).
- Square: a^2, 4a, diagonal a√2.
- Triangle: ½bh; equilateral (√3/4)a^2; Heron √(s(s−a)(s−b)(s−c)); right triangle ½ × legs.
- Parallelogram: base × height. Rhombus: ½ d_1 d_2. Trapezium: ½(a+b)h.
- Circle: πr², circumference 2πr; sector area (θ/360)πr²; arc (θ/360)2πr.
- Regular hexagon: (3√3/2)a^2.

Solids

- Cube: V = a^3, TSA 6a^2, diagonal a√3.
- Cuboid: V = lbh, TSA 2(lb+bh+hl), diagonal √(l^2+b^2+h^2).
- Cylinder: V = πr²h, CSA 2πrh, TSA 2πr(r+h).
- Cone: V = ⅓πr²h, slant l = √(r^2+h^2), CSA πrl, TSA πr(r+l).
- Sphere: V = 4/3 πr³, SA 4πr². Hemisphere: V = 2/3 πr³, TSA 3πr².
- Melting and recasting: **volume stays the same**.

Geometry facts

- Triangle angles sum to 180°; exterior angle = sum of the two opposite interior angles.
- Polygon with n sides: sum of interior angles = (n−2)×180°; each angle of a regular polygon = (n−2)×180/n.
- Pythagorean triples: (3,4,5), (5,12,13), (8,15,17), (7,24,25) and their multiples.
- Inscribed angle = half the central angle; angle in a semicircle = 90°.
- Similar triangles: sides ∝, areas ∝ square of sides.
- Number of diagonals of an n-gon = n(n−3)/2.



Circle inside a square, square inside a circle.

Find the area of a triangle with sides 13, 14, 15.

84.0

s = 21. Area = √(21×8×7×6) = √7056 = 84.

A cylinder has radius 7 cm and height 10 cm. Find its volume (π = 22/7).

1540

$V = \pi r^2 h = 22/7 \times 49 \times 10 = 1540$ cm³.

A metal sphere of radius 6 cm is melted and recast into small cones of radius 3 cm and height 4 cm. How many cones are formed?

24

Volume sphere = 4/3 π 216 = 288π. Cone = ⅓ π 9 × 4 = 12π. Number = 288/12 = 24.

The sides of a square are increased by 10%. By what % does the area increase?

21

$1.1^2 = 1.21$ → 21% increase.

Find the number of diagonals of a regular decagon, and the interior angle of a regular octagon.

35; 135°

Decagon: 10×7/2 = 35 diagonals. Octagon: 6×180/8 = 135°.

Find the length of a diagonal of a cuboid 4 × 3 × 12.

13

√(16 + 9 + 144) = √169 = 13.

The ratio of the areas of two circles is 4 : 9. Find the ratio of their circumferences.

2 : 3

Area ∝ r^2, circumference ∝ r. Ratio = √4 : √9 = 2 : 3.

A room 8 m × 6 m × 4 m is to be painted on four walls. Find the area (excluding a door of 6 m^2).

106

Walls = 2(l+b)h = 2×14×4 = 112 m^2. Minus door = 106 m^2.

Pattern	Wording	Method
Cost of paint / tiles	"area × rate"	Compute the right area first.
Melting & recasting	"melted and recast"	Equate volumes.
% change in area	"side increased by x%"	$(1+x)^2 - 1$.
Inscribed figures	"largest circle in square"	r = a/2.
Polygons	"interior angle of regular …"	(n−2)180/n.
Right triangle	"two sides 5, 12"	Triples.
Water in tank	"flows at … into a tank"	Volume = rate × time.

Shortcuts and tricks

- Recognise Pythagorean triples instantly.
- Ratio rule: area ∝ length², volume ∝ length³.
- Use π = 22/7 when the radius is a multiple of 7.
- "Largest" figure inside another always touches all the sides.

Common mistakes

- Using diameter as radius.
- Forgetting to square for area or cube for volume.
- Using CSA when TSA is asked (cylinder, cone).
- Mixing cm and m.

Practice set: Geometry practice

1. Area of a circle with radius 14 cm (π = 22/7)? **Basic**
2. Perimeter of a rectangle 12 × 8? **Basic**
3. Volume of a cube of side 5? **Basic**

4.	Find the area of a rhombus with diagonals 16 and 12.	Medium
5.	Largest circle is cut from a square of side 14 cm. Find its area (π = 22/7).	Medium
6.	The radius of a cone is 3 and its height is 4. Find the curved surface area in terms of π.	Medium
7.	Each interior angle of a regular polygon is 140°. How many sides?	Advanced
8.	A rectangular tank 6 m × 4 m is filled by a pipe of cross-section 0.5 m² at 2 m/s. How long to reach a depth of 3 m?	Advanced
9.	The perimeter of a rectangle is 56 and its area is 192. Find its diagonal.	Advanced
10.	A wire 88 cm long is bent into a circle then into a square. Find the side of the square.	Advanced

ADVANCED QUANT · TRICKS AND REVISION

15 Calculation Tricks, Mini Test and Formula Sheet

Speed decides the score. This chapter gives you the mental-maths tricks that save 20–30 seconds per question, a one-page formula sheet for last-day revision, and a timed mini test to check where you stand.

Squares and cubes you should know

n	n^2	n^3	n	n^2	n^3
1	1	1	14	196	2744
2	4	8	15	225	3375
3	9	27	16	256	4096
4	16	64	17	289	4913
5	25	125	18	324	5832
6	36	216	19	361	6859
7	49	343	20	400	8000
8	64	512	21	441	9261
9	81	729	22	484	10648
10	100	1000	23	529	12167
11	121	1331	24	576	13824
12	144	1728	25	625	15625
13	169	2197	26	676	17576

Memorise these. They appear inside almost every question.

Speed tricks

- **Square ending in 5:** $(n5)^2 = n(n+1)$ followed by 25. Example $75^2 = 7\times8 \mid 25 = 5625$.
- **Multiply by 11:** add neighbouring digits. $35 \times 11 = 3 \mid 8 \mid 5 = 385$.
- **Square near 100/50:** $98^2 = (100 - 2)^2 = 9604$; $52^2 = (50+2)^2 = 2704$.
- **$(a+b)(a-b) = a^2 - b^2$:** $47 \times 53 = 50^2 - 3^2 = 2491$.
- **Percent ↔ fraction:** 12.5 = 1/8, 16.67 = 1/6, 14.28 = 1/7, 11.11 = 1/9, 6.25 = 1/16.
- **Multiply by 5:** halve and ×10. $48 \times 5 = 240$. **By 25:** ÷4 ×100. **By 125:** ÷8 ×1000.
- **Divisibility:** 3 and 9 by digit sum, 4 by last two digits, 8 by last three, 11 by alternating sum, 7 by splitting 1001.
- **Approximation:** round numbers to 2 digits, compare options, and refine only if two options are close.
- **Option elimination:** use the last digit or the sign of the answer to remove options before calculating.

Compute 75^2 and 85^2.

5625; 7225

n(n+1) | 25: 7×8 = 56 → 5625; 8×9 = 72 → 7225.

Compute 47 × 53 without a calculator.

2491

(50 − 3)(50 + 3) = 2500 − 9 = 2491.

Find 17.5% of 480 quickly.

84

10% = 48, 5% = 24, 2.5% = 12. Sum = 84.

Estimate 4999 × 21 using the nearest round number.

104979

5000 × 21 = 105000 minus 21 = 104979.

One-page formula sheet

Topic	Must-know formulas
Number theory	Number of factors = ∏(e+1); zeros in n! = ⌊n/5⌋+⌊n/25⌋+...; HCF × LCM = product of two numbers.
Remainders	Cyclicity of last digit (4 for 2,3,7,8); Fermat; CRT; digital root mod 9.
Algebra	Sum of roots = −b/a; product = c/a; AM ≥ GM; discriminant $b^2 - 4ac$.
Series/logs	AP: n/2(2a+(n−1)d); GP: $a(r^n-1)/(r-1)$; log(ab) = log a + log b.
P&C	nPr = n!/(n−r)!; nCr = n!/(r!(n−r)!); circular (n−1)!; derangement D_n.
Probability	P(A∪B) = P(A) + P(B) − P(A∩B); P(A|B) = P(A∩B)/P(B); Bayes.
Sets	Inclusion–exclusion with 3 sets; exactly-two = pairs − 3·triple.
DI	% change = (new−old)/old; CAGR = (end/start)^(1/n) − 1.
Work	Total = LCM; rate = total/days; pipes: inlet +, outlet −.
TSD	Avg speed = 2ab/(a+b); relative speed; boats u±v; × 5/18.
Percent/P&L	Successive: a + b + ab/100; equal SP gain/loss ⇒ loss $x^2/100$.
Interest	CI = $P(1+r/100)^n - P$; CI − SI (2 yr) = $P(r/100)^2$.
Mensuration	Triangle ½bh; circle πr^2; cylinder $\pi r^2 h$; cone $\frac{1}{3}\pi r^2 h$; sphere $4/3\pi r^3$.

Timed mini test: 12 questions, 20 minutes

Practice set: Mini test (set a 20-minute timer)

1. Number of factors of 360? — Basic
2. Trailing zeros in 100!? — Basic
3. Remainder when 2^100 is divided by 7? — Medium
4. Sum of the first 20 terms of 3, 7, 11, ...? — Medium
5. Number of ways to arrange the letters of LEVEL? — Medium
6. Probability of two heads in three tosses? — Medium

7.	A job takes A 12 days and B 20 days. Together?	Medium
8.	Average speed for 60 km/h up and 40 km/h down?	Medium
9.	CI on ₹5,000 at 10% for 2 years?	Medium
10.	Marked 30% above cost, 10% discount: profit %?	Medium
11.	Mix ₹30/kg and ₹50/kg to get ₹42/kg: ratio?	Advanced
12.	Volume of a cone of radius 7 and height 6 (π = 22/7)?	Advanced

Shortcuts and tricks

- Do the questions you know first; leave the long ones for the end.
- Score yourself: 10+ is excellent, 7–9 is good, below 7 revisit the weak topic chapters.

Answer key

Answers to every practice set. B = Basic, M = Medium, A = Advanced.

1. How the section works

1. 24 (B)
2. -4 (B)
3. 80 (B)
4. 45 (B)
5. 22.5 (M)
6. 47 (M)

2. Number theory I

1. 48 (B)
2. 217 (B)
3. 12 (B)
4. 108 (B)
5. 37 (M)
6. 27 (M)
7. 12 (M)
8. 16 (A)
9. 200 (A)
10. 180 (A)

3. Remainders

1. 3 (B)
2. 5 (B)
3. 4 (M)
4. 1 (M)
5. 43 (M)
6. 11111111 (B)
7. 5 (M)
8. 58 (A)
9. 16 (A)
10. 1 (A)

4. Algebra

1. 26 (B)
2. 7 (B)
3. 4.5 (B)
4. 11 (M)
5. 4 (M)
6. 3 (M)
7. 3 (M)
8. 8 (A)
9. 4 (A)
10. 4 (A)

5. Series and logs

1. 1275 (B)
2. 670 (B)
3. 3025 (B)
4. 39366 (M)
5. 49/50 (M)
6. 35 (M)
7. 11 (A)
8. 2 (A)
9. 4 (A)
10. $\sqrt{5} + \sqrt{2} \approx 3.65$ (A)

6. Permutation and Combination

1. 60 (B)
2. 120 (B)
3. 66 (B)
4. 9 (B)
5. 120 (M)
6. 648 (M)
7. 63 (M)
8. 91 (M)
9. 120 (A)
10. 9 (A)
11. 84 (A)

7. Probability

1. 1/3 (B)
2. 1/2 (B)
3. 1/26 (B)
4. 5/14 (M)
5. 5/108 (M)
6. 5/16 (M)
7. 19/49 (A)
8. 2/3 (A)
9. 3/5 (A)
10. 33/100 (A)

8. Sets and Venn

1. 45 (B)
2. 13 (B)
3. 33 (M)
4. 30 (M)
5. 60 (M)
6. 16 (M)
7. 16 (M)
8. 734 (A)
9. 266 (A)
10. 11 (A)

9. Data Interpretation

1. 125% (B)
2. 560 (B)
3. Beta (B)
4. 10% (M)
5. 5:3 (M)
6. 123.75 (M)
7. 36000 (M)
8. 43.2° (M)
9. 2024 (A)
10. 1.2247 (A)

10. Time and Work

1. 12 (B)
2. 18 (B)
3. 28/5 (M)
4. 5 (M)
5. 10 (M)
6. 24/5 (A)
7. 12 (A)
8. 12 (A)
9. 12 (A)
10. 28 (A)

11. Time, Speed and Distance

1. 20 (B)
2. 72 (B)
3. 24 (M)
4. 5 (M)
5. 50 (M)
6. 20 (M)
7. 180 (A)
8. 4 (A)
9. 24 (A)
10. 36 (A)

12. Percentages, P&L, Mixtures

1. 0 (B)
2. 20 (B)
3. 19 (B)
4. 4 : 1 (M)
5. -2 (M)
6. 729/10 (M)
7. 4000 (A)
8. 20 (A)
9. 400 (A)
10. 1 : 1 (A)

13. Interest

1. 1200 (B)
2. 840 (B)
3. 13230 (B)
4. 25 (M)
5. 20 (M)
6. 820 (M)
7. 16 (A)
8. 900 (A)
9. 10000 (A)
10. 10 (A)

14. Mensuration and Geometry

1. 616 (B)
2. 40 (B)
3. 125 (B)
4. 96 (M)
5. 154 (M)
6. 15π (M)
7. 9 (A)
8. 72 (A)
9. 20 (A)
10. 22 (A)

15. Tricks, Mini Test and Formula Sheet

1. 24 (B)
2. 24 (B)
3. 2 (M)
4. 820 (M)
5. 30 (M)
6. 3/8 (M)
7. 15/2 (M)
8. 48 (M)
9. 1050 (M)
10. 17 (M)
11. 2 : 3 (A)
12. 308 (A)